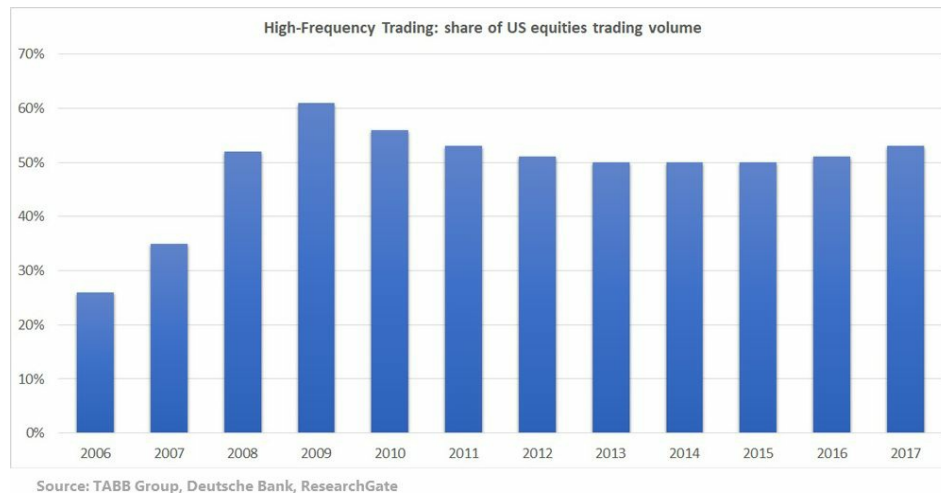




Although in Europe it is slightly lower, the share of the American equity market is still over 50% of the total volume traded. Interesting to note how the 2009 crisis caused a decline in HFT participation mainly due to increased competitiveness, high costs and low volatility.

Do not confuse High Frequency Trading with the automatic systems that can be created by a retail trader (which can fall into the category of algorithmic trading). Generally these types of tools (known as EAs, robots or bots) are not usually very effective; something very different from High Frequency Trading, which cost millions of dollars and have been developed by large financial firms to operate with significant amounts on a daily basis.

HOW DO HIGH FREQUENCY ALGORITHMS AFFECT US?

The fact that in today's markets most of the traded volume comes from high frequency algorithms does not greatly influence the structure-based analysis that we can do mainly because we are not competing to exploit the same anomalies.

While our analyses seek to take advantage of a deterministic side of the market where we try to elucidate who has more control (buyers or sellers), high frequency algorithms seem to be more in the random side of the market, mainly because of their categorization: arbitrage, directional (momentum and event-driven) and market making (liquidity ratio) strategies.

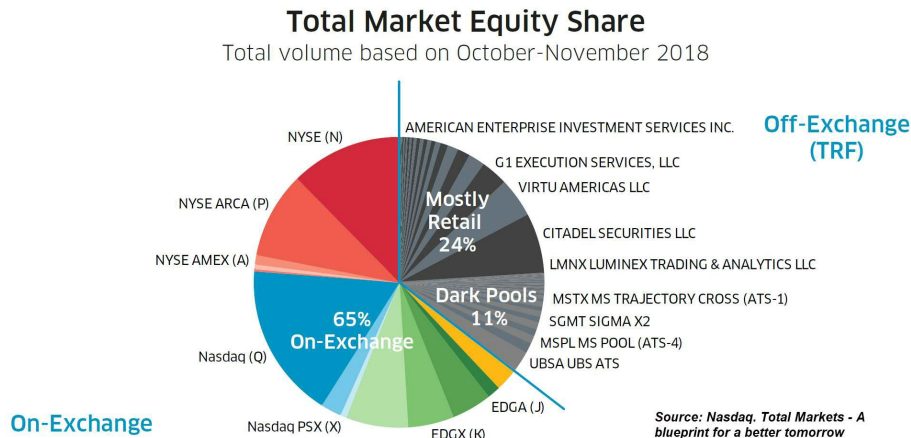
While it is true that some algorithms may execute directional strategies (with the objective of benefiting from price movements), these cover the shorter term and although they may distort our analysis, the advantage offered by the Wyckoff methodology is that it provides us with a structural framework thanks to which we can minimize some of the noise that originates on the smaller scales and obtain a more objective feeling of the current market conditions taking into account a larger context than that covered by these algorithms.

3.3 OVER THE COUNTER (OTC) MARKETS

This is a type of electronic market where financial assets are traded between two parties without the control and supervision of a regulator, as it is the case with stock exchanges and futures markets.

The main difference between centralized markets (On-Exchange) and non-centralized markets (Off-Exchange) is that in centralized markets there is a single order book which is responsible for linking all the participants in that market; whereas in non-centralized markets there are multiple order books (as many as there are market makers) where the lack of transparency with respect to market depth is evident by only showing the price of the BID and the ASK.

In recent years the US market has undergone a process of fragmentation where more and more decentralized markets have been created. Currently, U.S. equity liquidity is divided among about 88 different sources, with almost 40% of trading taking place in non-centralized markets.



Within the non-centralized markets we find different brokers based on their treatment of client orders. On the one hand, there are those who have a Dealing Desk (known as Market Makers) and, on the other hand, those who do not have a Dealing Desk (Non Dealing Desk), who act as intermediaries between the client and the rest of the market.

This second type, non-dealing desk brokers, is the one with whom it is recommended to work. The reason is because Market Makers are in charge of offering the final price of the asset, making the process less transparent.

Having the ability to take the counterpart in the trades of their clients opens the door to possible conflicts of interest, because if the client wins the broker loses and vice versa. And obviously the broker will do everything possible to keep his business profitable.

When the fact that the owner of the market is in charge of offering the final price together with the possibility that he is at the same time the counterparty, one of the main dangers for the retail trader appears and that is that he is exposed to suffer some kind of manipulation in the price movements.

It is also important to know that, due to the very nature of this type

of non-centralized markets, there may be different prices for the same asset. In other words, if we want to trade the EUR/USD currency cross, each market maker will offer us a different price and volume.

HOW DO OTC MARKETS AFFECT US?

The problem we encounter here is that the analysis we do under this type of market will be based on data that, although it could be a meaningful and valid representation of the whole market, in reality it does not genuinely represent all the price and volume data.

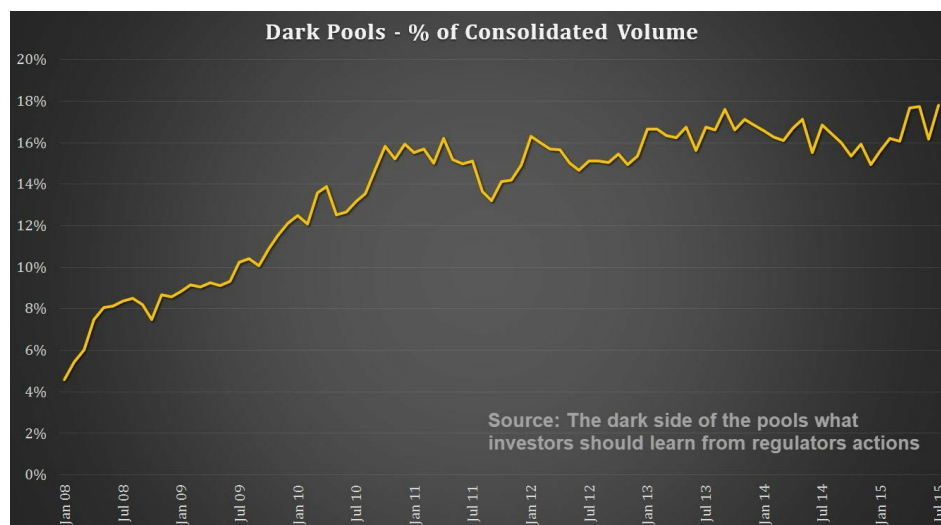
In order to reliably have this data we will need to analyze the asset under a centralized market. Continuing with the EUR/USD example, we should analyze the futures market (centralized market) which corresponds to the ticker \$6E.

Therefore, if we do not have the economic capacity (sufficient capital) to operate this futures market, we can analyze the asset in this futures market and execute the trading through another more affordable financial derivative such as the CFD (Contract For Different) with a good broker (which is not a market maker). An intermediate option would be to trade the small version of the future, the micro future, which in the case of EUR/USD corresponds to the ticker \$M6E.

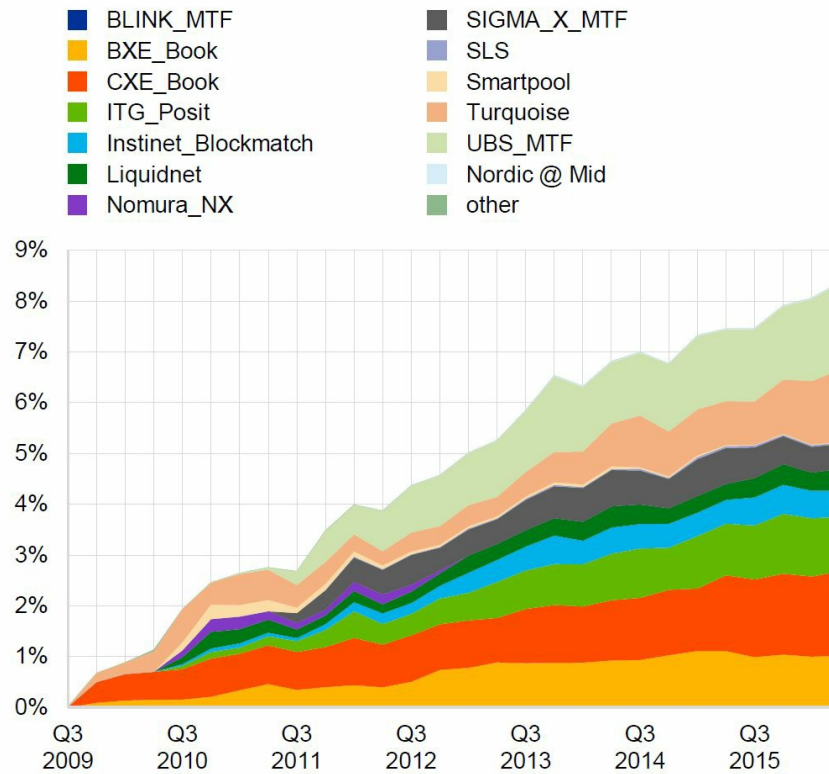
If we open a chart of the future (6E) and the CFD (EURUSD) we will see that the price movements are practically the same even though they are different markets. This is possible thanks to an arbitrage process carried out by high-frequency algorithms that occurs systematically between both markets.

3.4 DARK POOLS

A Dark Pool is a private market (Off-Exchange) that puts institutional investors in contact and facilitates the exchange of financial assets with the peculiarity that its transactions are not reported immediately, the amount traded (the volume) not being known until 24 hours later.



Trading in non-centralized stock markets in the U.S. is approximately 35%, with Dark Pools accounting for 16 to 18%. And according to a Bloomberg study, overall Dark Pools trading already accounts for over 30% of total trading volume.



Source: Dark pools in European equity markets: emergence, competition and implications

Dark Pools' market share of European equity trading has expanded rapidly in recent years, growing from 1% in 2009 to 8% in 2016.

When a large institution wants to buy or sell a huge amount of an asset they go to this type of market mainly because they know that if they access the public market they will find it difficult to find a counterparty, that they will possibly get a worse price in addition to being exposed to predatory techniques such as Front Running executed by high frequency algorithms. In this type of markets they avoid this negative impact and at the same time they get better commissions since they save the fees required by the public markets.

Contrary to what many people may think, Dark Pools are highly regulated as their owners are registered with the SEC (Securities and Exchange Commission) and FINRA (The Financial Industry Regulatory

Authority) and are therefore subject to regular audits and examinations similar to those of a public market.

In addition to private financial institutions, there are public exchanges that have their own Dark Pools such as the New York Stock Exchange (NYSE), the most traded and liquid exchange in the world.

The CME (Chicago Mercantile Exchange), which is the market with the largest number of options and futures contracts in the world, also has its own Dark Pool and offers this opaque trading service through what they have called "Block Trades". On their own website they explain in detail the information in this regard, to highlight:

“A Block Trade is a privately traded futures, options or combinations transaction that is allowed to be executed apart from the public auction market. Participation in Block Trades is restricted to participants in eligible contracts, as that term is defined in the Commodity Exchange Act.

Rule 526 ("Block Trades") governs Block Trades trading on the CME, CBOT, NYMEX and COMEX products. Block Trades are permitted in specific products and are subject to minimum transaction size requirements that vary by product, transaction type and execution time. Block Trades may be executed at any time at a fair and reasonable price".

HOW DO DARK POOLS AFFECT US?

The activity conducted in Dark Pools have an important role in determining intraday returns and the uncertainty that can be related to

them, thus having important microstructural implications.

It turns out that we may be analyzing an asset in which very significant transactions may have taken place in a hidden manner and we obviously cannot even assess the buyer's intentions.

These transactions, not being determined by the supply and demand of the public market, do not have an immediate impact on the price formation; but there are studies that affirm that the operators of the public markets react to the report of orders executed in the Dark Pool once it is released, being able to significantly alter the analysis of the interaction up to that moment.

3.5 ARE MARKETS RANDOM OR DETERMINISTIC?

This issue is another of the great debates within the trading community and undoubtedly generates great controversy. The vast majority of those who position themselves in favor of the randomness of the market do so with the aim of discrediting the usefulness of technical analysis. On the other hand, we have those who observe each and every price movement and give it an intention, a big mistake. Not everything is black and white.

Randomness is based on the premise of market efficiency while determinism (non-randomness) is based on market inefficiency.

The random markets approach suggests that the current price already reflects all the information about past events and even the events that the market expects to take place in the future. In other words, all the information about the asset is absolutely discounted and therefore it is not possible to predict the future price action. The reasoning is that, when participants try to take advantage of new information, together they neutralize this advantage. This would lead to the conclusion that it is not possible to take advantage of the market's own interpretation, unless the trader has access to privileged information.

The deterministic market approach suggests that price movements are influenced by external factors so that by knowing what these factors are, it is possible to predict future price action and therefore, it is possible to profit from the market's interpretation.

Randomness refers to the fact that this market movement has no logical intention behind it; it is simply a price fluctuation. Randomness arises as a result of the innumerable variables that take place in the market. No one can know how the other market participants are going to act. If anyone did, they would have a deterministic system whose predictions would be right every time.

On the other hand, if the Efficient Markets Hypothesis (EMH) and the randomness of the market were valid, no one could make profits on a recurring basis. And it has been proven throughout history that this is not the case. We all know great traders in the financial markets who have managed to win with different approaches (technical, fundamental and quantitative). In addition to this fact, the hypothesis of efficient markets has come in for a lot of criticism mainly because it assumes rational behavior of agents in their decision making.

On the other hand, financial markets cannot be modeled as a totally deterministic process where there is no randomness, since this would mean that there would be strategies with a 100% probability of success, and this (as far as we know) is not the case either.

3.5.1 THE ADAPTIVE MARKETS HYPOTHESIS

We therefore conclude that financial markets are composed of a percentage of randomness and another percentage of determinism, without knowing how much weight each has.

This theory would be supported by the Adaptive Markets Hypothesis (AMH) which shows the efficiency of financial markets not as a present or absent characteristic, but as a quality that varies according to the market conditions (the environment, context), which are determined by the interactions between its agents.

This hypothesis has been presented by the American financial economist Andrew W. Lo in his book *Adaptive Markets* published in 2017 and is mainly supported by:

1. **The efficiency of the market depends on its conditions.** This changing characteristic is the result of participants' interactions which in turn depend on market conditions.
2. **The agent is not totally rational and is subject to cognitive biases.** A purely rational model cannot be applied since participants form expectations based on different factors. Moreover, different expectations can be created with the same information, not to mention the fact that each agent has different degrees of risk aversion.

Although the author refers to agents as personal individuals, this is equally applicable to the current trading ecosystem where we have already mentioned that practically all actions are made electronically by algorithms. This fact does not modify the basis of the adaptive hypothesis since, regardless of the market participant and the way in which he interacts with the rest of the market, he will make his decisions based on the valuations, motives or needs that he has at a given moment; and that

given moment will be conditioned by different factors; factors that will change over time and thus change the valuations, motives or needs of the participants.

The HMA does not focus on discrediting the Efficient Markets Hypothesis, it simply treats it as incomplete. It places more value on changing market conditions (due to the arrival of new information) and how participants may react to them. It focuses on the fact that rationality and irrationality (efficiency and inefficiency) can coexist at the same time in the market depending on the conditions.

3.5.2 WHERE DOES THE WYCKOFF METHODOLOGY FIT IN?

Landing it on what concerns us, the reading of the market under the principles of the Wyckoff methodology is based on a deterministic market event: the law of Cause and Effect and for the market to develop an effect (trend) there must first be a cause (accumulation/distribution). There are other deterministic events that can offer an advantage, such as seasonality.

An example of random behavior could be seen in High Frequency Algorithms. We have already discussed some of their uses and they are the perfect example of forces that have the ability to move the market and that do not necessarily have a directional logic behind them.

Finally, most studies that defend the randomness of the market use classic chartist patterns such as triangles, shoulder head shoulder, flags

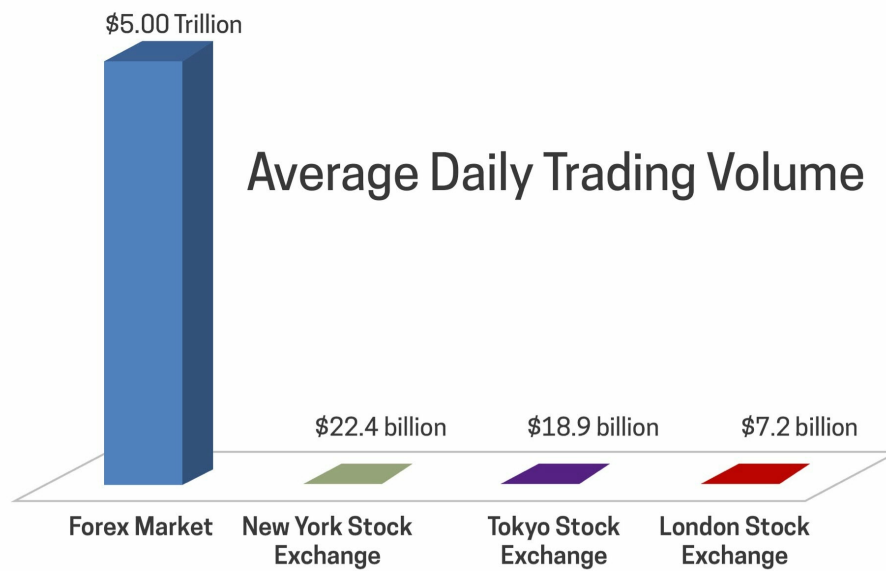
etc. or some price pattern without an underlying logic behind it to confirm the lack of predictability of technical analysis in general. Our approach to trading the markets is far removed from all this.

There are studies where using an analytical tool as simple as trend lines has shown a non-random behavior in financial markets being able to even exploit an anomaly with which to obtain certain returns.

PART 4. THE IMPORTANCE OF VOLUME

As we have seen, in the context in which we currently find ourselves, volume has become even more significant than in past decades. There is more and more money moving through all the financial markets and this has caused certain changes; from the way of trading to the appearance of new tools.

At the beginning of the 20th century, the markets, operated entirely manually, were guided mainly by the cognitive biases of their participants. Emotions such as fear and greed were present and caused a large part of the decisions made by their participants. This irrationality of the individual led to very profitable situations for the well-informed traders of the time.



As everyone knows by now, the current ecosystem is that the vast majority of traded volume is done electronically, that huge amounts of volume are moved on a daily basis and that in order for all these orders to be filled, it is necessary to highlight the concepts of counterparty, liquidity and order matching; in short, the importance of volume.

In this section we will study auction theory in depth and look at some tools that will allow us to make a more accurate analysis of volume data.

4.1 AUCTION MARKET THEORY

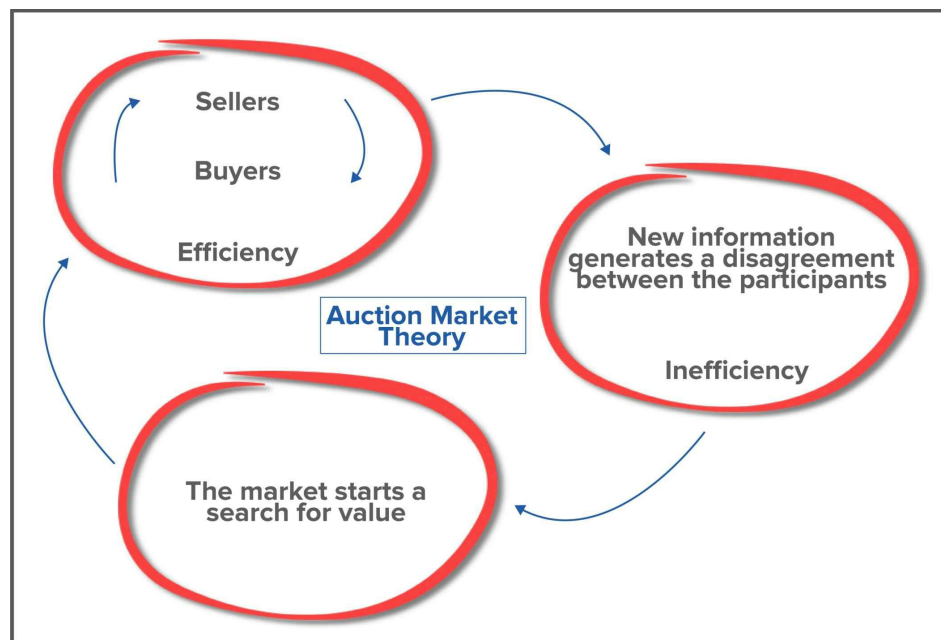
Auction theory, as it is known in English, was born mainly from the studies by J.P. Steidlmayer on the Market Profile. Subsequently, together with other authors such as James Dalton and Donald L. Jones, he defined a series of concepts that constitute this theory.

It is based on the fact that the market, with the priority objective of **facilitating negotiation** among its participants and under the principles of the law of supply and demand, will always move in search of efficiency, also known as equilibrium or fair value.

Efficiency means that buyers and sellers are comfortable negotiating and neither has clear control. That comfort comes because, based on current market conditions, the valuations of both are very similar. The way this equilibrium is visually observed on a price chart is with a continuous rotation (price ranges). This price lateralization represents this equilibrium. It is evidence of the facilitation of trading and is the state in which the market is always looking to be.

On the other hand, we have the moments of **inefficiency** or imbalance and these are represented in the trend movements. When new information reaches the market, it can cause the value perceived by both

buyers and sellers of the asset to change, generating a disagreement between them. One of the two will take control and move the price away from the previous equilibrium zone, offering a profitable trading opportunity. What is evident in this context is that the market is not facilitating trading and is therefore considered an inefficient condition.



The market will be constantly moving in search and confirmation of value; in situations where buyers and sellers are in a position to exchange stock. When this happens it is because the valuations that these participants have on the price are very similar. At that moment the negotiation will again generate a new equilibrium zone. This cycle will repeat itself over and over again in an uninterrupted manner.

The general idea is that the market will move from one **equilibrium** area to another through trend movements and that these will be initiated when the market sentiment of both buyers and sellers about the current value differs causing the **imbalance**. The market will now begin the search for the next area that generates consensus among the majority of participants about the security.

It should be noted that the market spends most of its time during periods of equilibrium, which is logical due to the nature of the market based on favoring negotiation among its participants. This is where the accumulation and distribution processes take place, which as we all know is where the Wyckoff methodology focuses.

4.1.1 VARIABLES

The auction process in the financial markets is fundamentally based on value. In order to try to decipher where such value is found, three additional elements need to be evaluated:

PRICE

In the auction mechanism price is used as a discovery tool. The facilitation of trading is carried out by price movement, which fluctuates up and down exploring different levels with the objective of seeing how participants react to such exploration.

These price movements herald opportunities. If the participants respond to this exploration by observing the price as fair, it will provoke negotiation between them. Conversely, if these discoveries of new price levels are not perceived as attractive to both participants, it will lead to rejection.

TIME

When the market promotes an opportunity (reaches an attractive level), it will use time to regulate the duration that the opportunity will be available.

The price will spend very little time in those zones that are advantageous to one of the two sides (buyers or sellers).

A zone of efficiency or equilibrium will be characterized by a higher consumption of time; while a zone of inefficiency or imbalance will be represented by a short consumption of time.

VOLUME

Volume represents the activity, the amount of an asset that has been exchanged. This quantity suggests the interest or disinterest at certain levels.

Based on volume there are areas that are more valuable than others. The basic rule is that the more activity a certain area sees, the higher the value assigned to it by market participants.

$$\textit{Price} + \textit{Time} + \textit{Volume} = \textit{Value}$$

These three elements are responsible for giving us a logical perspective on, based on current conditions, where market participants consider the value of a particular asset to be.

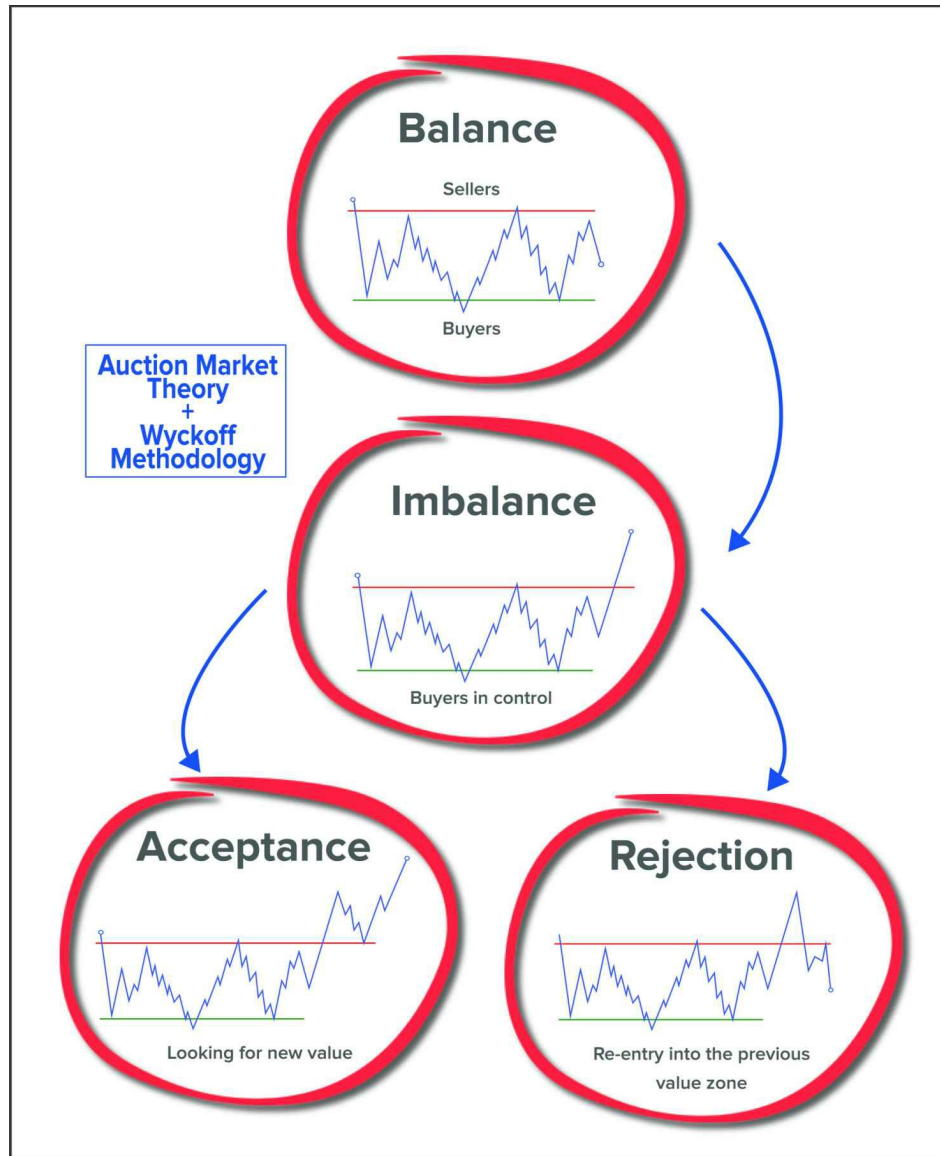
Through price the market discovers new levels, time consumption suggests that there is some acceptance in that new area and finally volume generation confirms that participants have created a new value zone where they trade comfortably.

As we know, conditions are changeable and therefore continuous re-evaluation of these elements is needed. Knowing where the value is a key as it defines the market condition and based on this we will be able to put forward different trading ideas.

4.1.2 PERCEPTION OF VALUE

The market is constantly rotating between two phases: horizontal development (equilibrium) or vertical development (imbalance). Horizontal development suggests agreement among participants while vertical development is a market in search of value, in search of participants to negotiate with.

The fact that the price is moving comfortably in a trading range (horizontal development) represents acceptance in that zone, it is a context where price and value coincide according to the participants. When the market is in a trending state (vertical development) price and value do not coincide; in this context price will move ahead and value will follow or not (as a sign of acceptance and rejection).



In an equilibrium area the fairest price will be located in the middle and the extremes both above and below will represent levels that are unfair or not accepted by the participants.

Since the fairest value is located in the middle of the range, a move to the upper end will be seen by buyers as an expensive price and at the same time sellers will consider it cheap, so their actions will lead to send the price back to the fairest zone. Similarly a visit to the lower end of the range will be seen as cheap by buyers and expensive by sellers, leading to

a further turn to the upside.

It is nothing more or less than the typical range trading that we all know where you look to buy at lows and sell at highs hoping that the price will continue to reject those extremes. And normally the market will continue to do so until its condition changes.

The interesting thing comes when an imbalance occurs and the price leaves the value zone.

What will happen then? When the price leaves a trading zone a change in the perception of value can happen.

The trader's task is now to assess whether these new price quote levels are accepted or rejected. Price is ahead of the other two variables in determining potential areas of value, but it is time in the first instance and volume in the last instance that will confirm whether that new area is accepted or rejected.

We interpret acceptance to a new area when the price manages to hold (time consuming) and contracts begin to be exchanged between buyers and sellers (volume), all this represented as a certain lateralization of the price. Conversely, we would identify rejection when the price quickly reverts back to its old value area denoting lack of interest and evidenced by a sharp turn.

All horizontal developments end when there is no longer agreement among the participants about the value; while all vertical developments end when the price reaches a zone where there is agreement among them again. This is the continuous cycle of the market. This idea in itself is very powerful and with the right approach trading strategies could be built around it.

As with one of the universal principles of technical analysis (price discounts everything), we do not need to go into what actually produces this change in the participants' perception of value. We know that based on the current conditions, based on the information available at that precise moment, all participants give a valuation to the price of the asset. Subsequently, something may happen at a fundamental level that changes this perception, but the beauty of this approach is that it removes us from the need to know and interpret what has happened to change the participants' perception.

It is important to emphasize that this auction theory is universal and can therefore be used to evaluate any type of financial market, regardless of the time frame used.

4.1.3 THE FOUR STEPS OF MARKET ACTIVITY

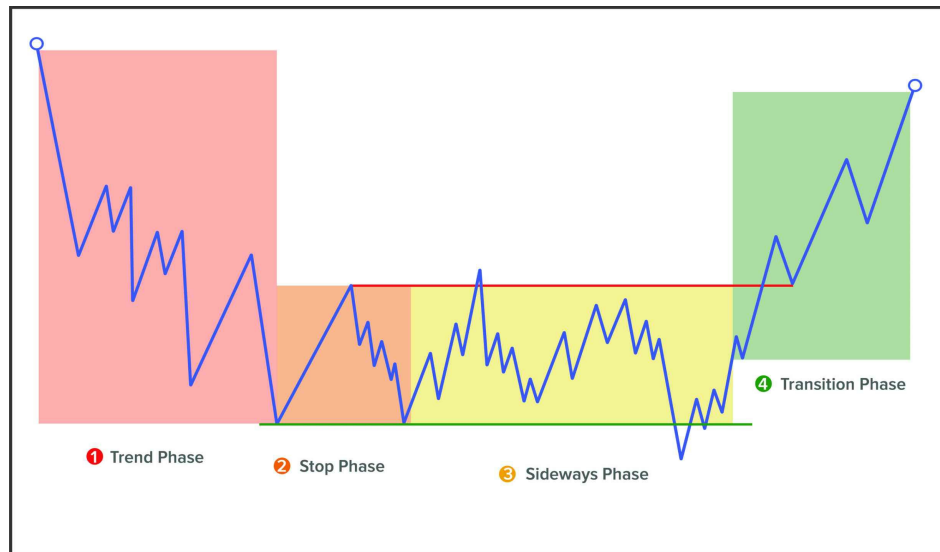
This is a process with which Steidlmayer represented the different phases through which the market passes during the development of its movements.

The four phases are:

1. **Trend phase.** Vertical development, price in imbalance in favor of one direction.
2. **Stopping phase.** Traders begin to appear in the opposite direction and the stop of the previous trend movement occurs.

The limits of the upper and lower range are established.

3. **Lateralization phase.** Horizontal development. Trading around the stop price and within the limits of the new equilibrium range.
4. **Transition phase.** The price leaves the range and a new imbalance begins in search of value. This movement may be a reversal or a continuation of the previous trend movement.



Once the transition phase is over, the market is in a position to start a new cycle. This protocol will run uninterrupted and is observable in all time frames.

Visually up to step three a P or b-shaped profile would be observed. The formation of this type of profile as well as the proposal of a type of operative based on it will be seen in more depth later.

Structural traders will be familiar with this four-step protocol because it is essentially exactly the development from Phase A to Phase E proposed by the Wyckoff methodology:

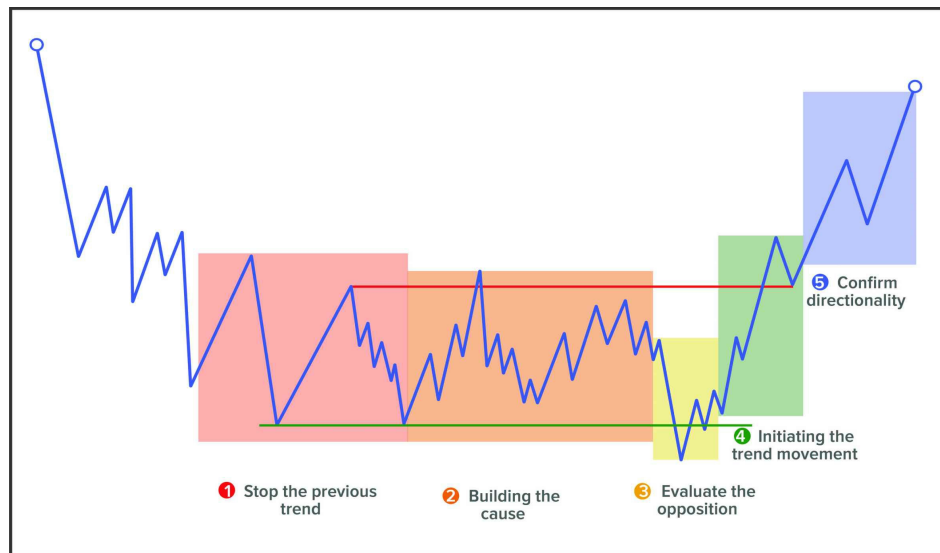
1. Stop the previous trend

2. Build the cause

3. Assess the opposition

4. Initiate the trend movement

5. Confirm directionality



Although Richard Wyckoff, as well as his later students and other traders who have contributed to the dissemination of his ideas were based solely on analytical tools and principles of technical analysis, we see that they already implicitly worked these concepts proposed in the auction theory even if they did not use such terms.

That is why it is considered the only technical analysis approach based on a real underlying logic: auction theory and the law of supply and demand.

4.2 THE LAW OF SUPPLY AND DEMAND

This is the basic law on which the auction theory is based and therefore governs all price changes. Initially the studies that Richard Wyckoff proposed on this law told us that:

- ❖ If demand was greater than supply, the price of the product would rise.
- ❖ If supply was greater than demand, the price of the product would go down.
- ❖ If supply and demand were in equilibrium, the price of the product would remain the same.

4.2.1 COMMON ERRORS OF INTERPRETATION

This idea is very general and should be qualified since a series of conceptual errors have been generated around this law of supply and

demand.

Error No. 1: Prices rise because there are more buyers than sellers or fall because there are more sellers than buyers.

In the market there are always the same number of buyers and sellers; since for someone to buy, there must be someone to sell to him. No matter how much someone wants to buy, as long as there is no seller willing to offer the counterpart, it is impossible for the negotiation to take place.

The key lies in the attitude (aggressive or passive) that traders take when participating in the market.

Error No. 2: Prices go up because there is more demand than supply or they go down because there is more supply than demand.

The problem with this statement is to call demand everything that has to do with buying and supply everything that has to do with selling. In reality they are different concepts and it is convenient to separate them when referring to one or the other.

Supply and Demand are the limit orders that both buyers and sellers place in the BID and ASK columns pending execution, which is known as liquidity.

4.2.2 BID/ASK, SPREAD AND LIQUIDITY

There is no single price in financial markets. This is obvious but not understood by many people. When a participant goes to the market, he finds two prices: the buying and the selling price.

BID. The BID column is the part of the order book where buyers come to place their bid (limit buy orders) and where sellers come to match their ask orders. The highest price level within the BID column is known as the Best BID and represents the best price at which to sell.

ASK. The ASK column is the part of the order book where sellers go to place their pending sell (bid) orders and where buyers go to find the counterpart for their purchases. The lowest price level within the ASK column is known as the Best ASK and represents the best price at which to buy.

The diagram illustrates an order book with three columns: BID, PRICE, and ASK. The BID column (green) lists buy limit orders with quantities. The PRICE column (grey) lists the corresponding prices. The ASK column (red) lists sell limit orders with quantities. The Best BID is the highest price in the BID column (100), and the Best ASK is the lowest price in the ASK column (75). The Last Crossing Price is the price at which the best bid and best ask meet (100). The Demand Buy Limit Orders are the orders in the BID column, and the Supply Sell Limit Orders are the orders in the ASK column.

BID	PRICE	ASK
	108	600
	107	980
	106	900
	105	720
	104	550
	103	500
	102	120
	101	90
	100	75
50	99	
66	98	
95	97	
130	96	
249	95	
120	94	
97	93	
90	92	

Therefore, it is the order execution mechanism that determines the price. The difference between the BID and the ASK is called the **Spread** and is an indicator to be taken into account to assess the liquidity of the

asset. The lower the Spread, the more liquid it is.

Liquidity is an extremely important concept. It is the amount of volume that an asset trades. We should try to trade assets the more liquid the better, as this will mean that it will be more difficult for a large individual trader to have the ability to move the price. It is therefore a measure to avoid possible manipulation. If you are trading an asset that trades very little volume, it is likely that if a large institution enters, it can move the price with relative ease. These environments should be avoided.

4.2.3 TYPES OF PARTICIPANTS BASED ON THEIR BEHAVIOR

The key element that clarifies the errors in the interpretation of the Law of Supply and Demand is the behavior of traders, since they can participate in the market in different ways:

- ❖ **Aggressive.** Liquidity takers through the use of "at market" orders. They have urgency to enter and attack the Best BID and Best ASK where the limit orders remain located. This type of aggressive orders is the real engine of the market since they are the ones that initiate the transactions.
- ❖ **Passive.** Liquidity creators through the use of "limit" orders. Sellers create supply by leaving their orders pending execution in the ASK column; and buyers create demand by leaving their orders placed in the BID column.

BID	PRICE	ASK
	108	600
	107	980
	106	900
	105	720
	104	550
	103	500
	102	120
	101	90
	100	75
50	99	
66	98	
95	97	
130	96	
249	95	
120	94	
97	93	
90	92	

Agressive Seller
Takes liquidity from the BID

Passive Buyer
Provides liquidity in the BID

Passive Seller
Provides liquidity in the ASK

Agressive Buyer
Takes liquidity from the ASK

4.2.4 HOW DOES PRICE MOVEMENT OCCUR?

We come then to the moment of examining what must happen for the price to move. The idea here is clear: it takes the aggressive participation of traders to produce a change in price. Passive orders represent intent in the first instance, and if executed they have the ability to stop a move, but not the ability to make the price move. This requires initiative.

INITIATIVE

For the price to move upwards, buyers have to buy all the sell (bid) orders that are available at that price level and also continue to buy aggressively to force the price up one level to find new sellers to trade with.

Passive buy orders cause the downward movement to slow down, but by themselves cannot drive the price up. The only orders that have the ability to move the price upwards are buy-to-market orders or those whose crossing orders become buy-to-market orders (such as stop losses on short positions).

BID	PRICE	ASK
	108	600
	107	980
	106	900
	105	720
	104	550
	103	500
	102	120
	101	90
	100	75
50	99	
66	98	
95	97	
130	96	
249	95	
120	94	
97	93	
90	92	

For the price to move down one level sellers must consume those 50 limit orders

For the price to move up one level buyers must consume those 75 limit orders

For the price to move downward, sellers have to acquire all the buy (demand) orders that are available at that price level and continue to push downward forcing the price to go in search of buyers at lower levels.

Passive sell orders cause the upward movement to slow down, but do not have the ability to drive the price down on their own. The only

orders that have the ability to move the price downward are market sells or those whose crossing orders become market sells (Stop Loss of long positions).

EXHAUSTION

The price needs aggressiveness to move, but it is very interesting to take into account also that the lack of interest from the opposite side can facilitate this task.

An absence of supply can facilitate the price rise just as an absence of demand can facilitate its fall.

BID	PRICE	ASK
	108	134
	107	100
	106	180
	105	120
	104	44
	103	26
	102	15
	101	7
	100	5
50	99	
66	98	
95	97	
130	96	
249	95	
120	94	
97	93	
90	92	

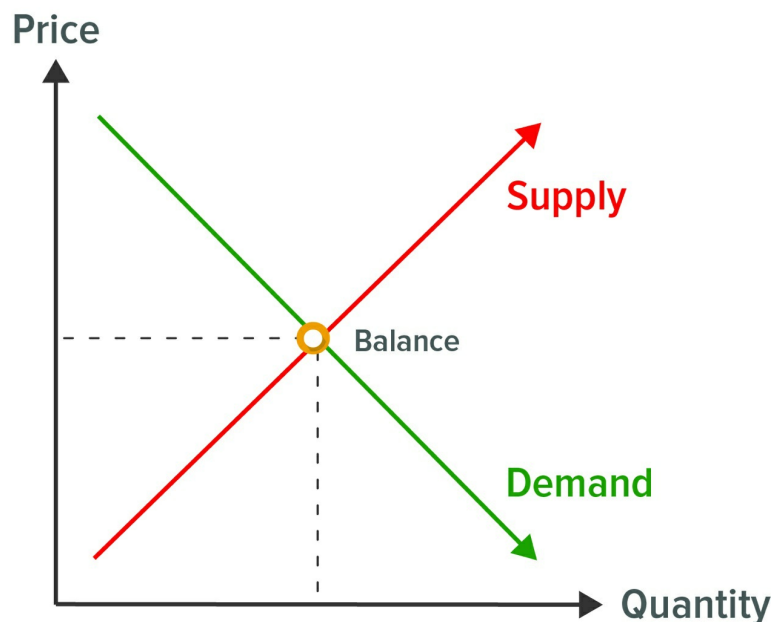
Large imbalance between buy and sell limit orders = Lack of selling interest

As supply withdraws, this lack of interest will be represented as fewer contracts placed in the ASK column and therefore the price can more easily move upward with very little buying power.

Conversely, if it is demand that withdraws, it will be visualized with a reduction in the contracts that buyers have placed in the BID and this will cause the price to be able to move lower with very little selling initiative.

4.2.5 HOW DO MARKET TURNS OCCUR?

Keep in mind that the market, in order to facilitate trading, will go up in search of sellers and down in search of buyers; in other words, it will always move towards the equilibrium point where supply and demand will be equalized.

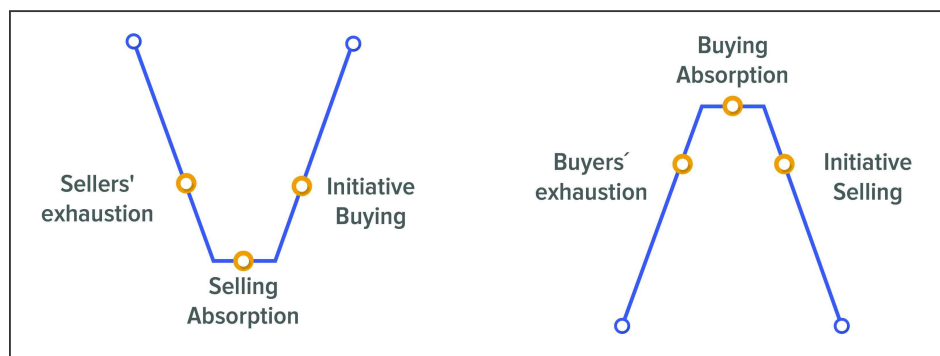


Moreover, logic leads us to think that as the price rises, the interest of buyers decreases (they see the price as increasingly more expensive) and the interest of sellers increases (they see the price as increasingly cheaper); and as the price falls, the interest of sellers decreases and that of buyers increases.

In a rising market, as long as the buying initiative is able to consume all the liquidity (supply) found at higher levels, the price will continue to rise. On the other hand, in bear markets, as long as the selling initiative is able to consume all the liquidity (demand) found at lower levels, the price will continue to fall.

At the moment of a market turn, we will normally always have a three-step process:

1. **Exhaustion**
2. **Absorption**
3. **Initiative**



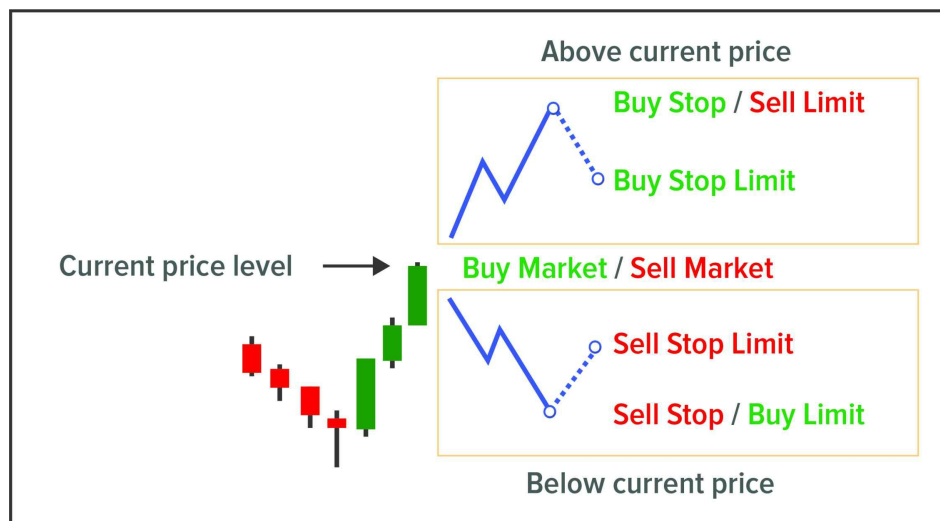
To reverse an upward movement, the lack of interest (exhaustion) of the buyers to continue buying, the first entry of sale by the big operators in a passive way (absorption) and the aggressiveness (initiative) of the sellers will be combined.

Oppositely for the bullish turn example: sellers' exhaustion, passive positioning through absorption of sales and buying initiative with aggression in the ASK.

In essence this three-step protocol is nothing more than accumulation and distribution processes regardless of the time scale on which they take place.

4.3 ORDER TYPES

When participating in the market we can do so mainly by using four different types of orders.



- ❖ **Market.** Aggressive order that is executed at the best available buy and sell price (Best BID/ASK). Immediate execution that guarantees the entry to the market but not the specific price at which it has been entered due to the constant change in the quotation and the application of the Spread.
- ❖ **Limit.** Passive order that is executed at a specific price. Entry at that specific price is guaranteed but execution is not guaranteed.

That is, the price may never visit that level and therefore we would not enter the market. As long as they are not executed, they can be removed at any time.

- ❖ **Stop.** A passive order that is executed at a specific price. When the price reaches it, it becomes a market order and is therefore executed at the best available price (Best BID/ASK).
- ❖ **Stop-Limit.** Combines the features of Limit and Stop orders. Once a certain price level is reached (Stop orders function), an order is generated at a specific level (Limit orders function). Operationally, it works exactly the same as Limit orders.

Based on the four types of orders we have just seen, the following is the full spectrum of basic orders available depending on the intention and what it is used for:

- ❖ **Buy Market.** Aggressive order at the current price. It is used to:
 - Enter the market on buy.
 - Close a sell position (either at a profit or loss).
- ❖ **Buy Stop.** Pending order above the current price. It is used to:
 - Enter the market on buy.
 - Close a sell position (by Stop Loss).
- ❖ **Buy Limit.** Pending order below the current price. It is used for:
 - Enter the market on buy.
 - Close a sell position (by Take Profit).
- ❖ **Buy Stop Limit.** Pending order below the price after reaching a

level. It is used to:

- Enter the market to buy.
- Close a sell position (by Take Profit).

❖ **Sell Market.** Aggressive order at the current price. It is used to:

- Enter the market on sell.
- Close a buy position (either in profit or loss).

❖ **Sell Stop.** Pending order below the current price. It is used to:

- Enter the market on sell.
- Close a buy position (by Stop Loss).

❖ **Sell Limit.** Pending order above the current price. It is used to:

- Enter the market on sell.
- Close a buy position (by Take Profit).

❖ **Sell Stop Limit.** Pending order above the price after reaching a level. It is used to:

- Enter the market to sell.
- Close a buy position (by Take Profit).

4.3.1 ADVANCED FEATURES

There are certain advanced instructions that can be applied, depending on the broker, to orders to enter and exit the market:

One-Cancels-Other (OCO). Entering two orders to the market with one of them being cancelled when the other is executed.

Order-Sends-Order (OSO). Instruction to execute secondary orders when the initial one is executed.

Market-To-Limit (MTL). Execution of a market order that includes an instruction to place a limit order at the same price with the remainder of the position if it is not fully executed.

Market-If-Touched (MIT). Execution of a market order that includes an instruction to place a limit order at the same price with the remainder of the position if it is not fully executed.

Limit-If-Touched (LIT). Conditional order to trade with a pending order (Limit) when a specific level is reached. It is used to buy above the current price and to sell below the current price.

Good-Til-Cancelled (GTC). It includes a temporary execution period which is usually the duration of the session. If at that point the order has not been executed, it will be cancelled.

Good-Til-Date (GTD). The order remains active until a specific date.

Immediate-Or-Cancel (IOC). Instruction to execute the order immediately. If any portion of the order remains unfilled, that portion is cancelled.

Fill-Or-Kill (FOK). It does not allow partial execution. When the price is reached, it either enters the market with full volume or is

cancelled.

All-Or-None (AON). Similar to the FOK orders with the difference that in case the price reaches it and it is not executed because it cannot cover all its volume, it remains active until it obtains all the counterpart or it is cancelled by the trader.

At-The-Opening (ATO). Instruction to execute an order at the opening of the session. If this is not possible, it is cancelled.

At-The-Close (ATC). Instruction to execute an order at the close of the session.

4.4 TOOLS FOR VOLUME ANALYSIS

Thanks to certain tools that analyze the flow of orders we can see all the interaction between buyers and sellers who participate in the market in different ways.

We will mainly distinguish them by the type of orders they deal with as not all these tools are based on the same data:

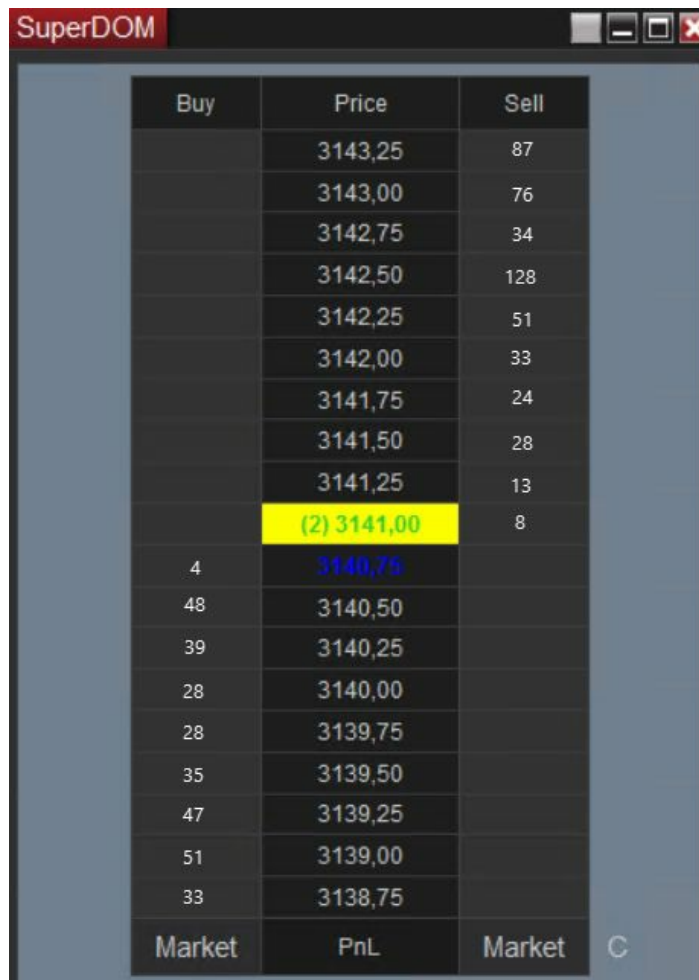
- ❖ Analysis of pending orders: Order Book, also called Depth of Market (DOM).
- ❖ Analysis of executed orders: Time & Sales and Footprint.

We will point out their most important characteristics in order to provide the reader with a basic knowledge of the peculiarities of each one.

4.4.1 ORDER BOOK

Identifies all pending orders for execution (liquidity) that are located in the BID and ASK columns; as we already know, the BID

represents pending buy orders and the ASK represents pending sell orders.



Buy	Price	Sell
	3143,25	87
	3143,00	76
	3142,75	34
	3142,50	128
	3142,25	51
	3142,00	33
	3141,75	24
	3141,50	28
	3141,25	13
	(2) 3141,00	8
4	3140,75	
48	3140,50	
39	3140,25	
28	3140,00	
28	3139,75	
35	3139,50	
47	3139,25	
51	3139,00	
33	3138,75	
Market	PnL	Market C

As these types of software only have Level II of depth, most markets can only show ten levels of liquidity above and below the current price. This is relevant because beyond those ten levels there will still be liquidity located, but it is only visible to those who have Level III of depth (mainly liquidity providers).

The analysis of the order book presents some problems. One of them is that the visible liquidity is never the liquidity that is actually located in those levels. The liquidity that can be seen in the order book comes only from limit orders. By their very nature, market orders cannot

be seen anywhere, since they are born from an initiative and are executed instantaneously. On the other hand, Stop orders, as they become a market order when their price is reached, are not displayed in the Order Book either.

In addition to this, as we have seen in the advanced features of some orders, there are certain instructions that are not displayed in the Order Book either, so that in reality we would be analyzing a not very high percentage of the total orders pending execution.

Another major problem with the analysis of the order book or any other tool based on this liquidity data is that these orders that are pending execution can be removed by whoever placed them there at any time before they are executed. Due to this peculiarity different forms of manipulation carried out by algorithms emerged:

SPOOFING

This is the placement of huge amounts of contracts in the BID and ASKS columns (limit orders) without the intention of actually executing them. The aim is to give the impression of an "insurmountable barrier" and cause the price to move to the opposite side. They are false orders because when the price is about to reach its level, they are cancelled and are not executed.

It is an interesting concept that highlights the value of the ability of limit orders with respect to price movement. As we have mentioned, limit orders by themselves and by their very nature do not have the ability to move the market, but using this form of manipulative activity we see how

the price can move influenced by them at certain times. Not directly based on their execution, but indirectly based on their influence.

Imagine that you normally see limit orders of around 50 contracts per price level. What will the other participants think if you suddenly see 500 contracts? Well, most logically they will see that this is a too high price to pay for the price to cross that level and most likely cause a lack of interest to go against such orders. And of course, this will result in a move in the opposite direction of the huge order. Manipulation by large traders to drive the price in the direction they want.

ICEBERG ORDERS

This is the splitting of a large limit order into smaller portions. The motivation for this type of action has to do with wanting to hide the actual size of the original order.

Mainly used by institutional traders who want to execute a huge number of contracts in a given price range and who use algorithms programmed with this technology to be able to do so passively and without turning the price against them. It is important to note that there is only one source behind such an order, just one large trader, not a collection of them.

It is very visual to take the example of a real Iceberg. On the surface what you see is a seemingly normal amount of contracts, but what you don't know is that this order is simply a part of a much, much larger one. And when this small part is filled by the market the large order quickly replenishes it.

This is the clearest example of what a **takeover** would look like. There may be many aggressive buyers pushing the ASK and all those Buy Markets are crossing the Sell Limits of an Iceberg order which do not allow the price to move up. We will say that a takeover of these purchases is taking place.

4.4.2 TAPE

Thanks to Time & Sales we can see in real time all the crossing of orders already executed. The analysis of the tape becomes very complex due to the speed at which the current markets move (mainly futures markets).

T & S			
Bid		Ask	
3141,50	9	3141,75	3
11:03:15	3141,75	1	
11:03:15	3141,75	2	
11:03:15	3141,75	1	
11:03:15	3141,75	1	
11:03:15	3141,75	1	
11:03:14	3141,75	1	
11:03:14	3141,75	1	
11:03:14	3141,75	1	
11:03:13	3141,75	2	
11:03:13	3141,75	1	
11:03:13	3141,75	2	
11:03:13	3142,00	1	
11:03:12	3142,00	1	
11:03:12	3141,75	1	
11:03:12	3141,75	1	
11:03:12	3141,75	1	
11:03:12	3141,75	1	
11:03:12	3141,75	1	
11:03:12	3141,75	1	
11:03:12	3141,75	1	
11:03:12	3141,75	2	
11:03:12	3141,75	1	
11:03:12	3141,75	1	
11:03:12	3142,00	2	
11:03:12	3142,00	1	
11:03:12	3142,00	2	
11:03:12	3142,25	1	
11:03:12	3142,00	1	
11:03:11	3142,00	1	
11:03:11	3142,00	1	
11:03:11	3142,00	1	

Depending on the software we can access different types of information. It usually includes at least the columns with the time of execution, the price level and the number of contracts.

It can certainly be useful for the identification of large volumes executed in a single order, known as "big trade". More modern versions also allow the grouping of orders executed in blocks, pointing to the same trader as the origin of different transactions spaced in time.

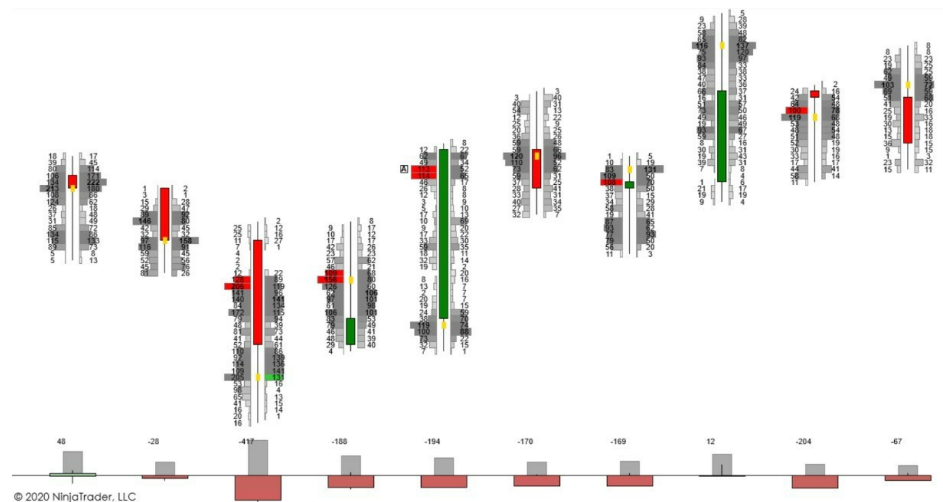
A positive point of the Tape with respect to the Order Book is that the tape represents the past, the orders already executed and therefore is

not susceptible to manipulation.

Operating solely by analyzing the tape is not within everyone's reach since it requires enormous experience as well as a great capacity for concentration due to the speed at which it moves.

4.4.3 FOOTPRINT

When most people refer to Order Flow they are actually referring to this section of the order flow, the use of Footprint. Order Flow is a general term that, as the name suggests, is concerned with order flow.



The Footprint charts the data provided by the tape (orders already executed) and represents them in a much more visual way. It would be like putting a magnifying glass inside the candlesticks and observing the number of contracts executed at each price level..

The advantage of analyzing the Footprint is that it allows us to

quantify in maximum detail the interaction between buyers and sellers. Observing the balance and imbalance between participants as well as being able to identify in which column more volume is being traded can certainly be useful at certain times.

There are different types of footprint charting based on:

- ❖ The nature of the data: it can be set up with time, range, volume and rotation candlesticks.
- ❖ The representation protocol: Profile, Delta, Imbalance, Histogram, Ladder or BID/ASK.

It is a very configurable tool that generally includes multiple functionalities although it basically analyzes the orders executed by price levels with the objective of looking for imbalances, takeovers, initiatives, unfinished auctions, clusters, big trades etc.

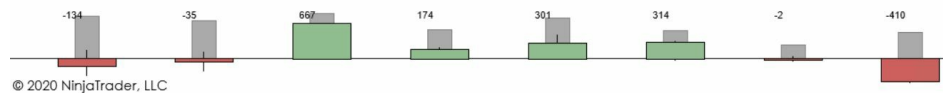
Once you know in depth how the order crossing is executed, you will come to the conclusion that its analysis is very subjective and that trading based on this tool without taking into account anything else is not advisable.

4.4.2 DELTA

A more specific and popular tool in the world of Order Flow analysis is the Delta. The delta is an indicator that simply measures the difference between the volume traded on the BID and the volume traded on the ASK in a given period of time. If the difference is positive the delta

will be positive and vice versa if it is negative. In addition, the difference between deltas can also be visualized since the indicators will normally show them with different sizes.

There is a common misconception that all volume traded on the ASK "are buys" and all volume crossed on the BID "are sells", although in reality with that statement they mean that they are directional in origin; that is, they have an intention to add pressure towards one of the sides. If this were so, why do we sometimes see bearish moves with positive deltas and bullish moves with negative deltas? Believe me it's not that simple. If it were, we would have found the Holy Grail. Again it has to do with the crossover mechanism.



For the time being, let's keep in mind that it is visually observed on a horizontal axis with value 0 and that depending on the column where the transactions are executed, it will be displayed positively or negatively on the axis. This representation would be for the example of the normal Delta since there is another variant, the accumulated, where it is charted continuously without taking into account the horizontal axis.

The delta is updated at each executed order and therefore gives rise to the possibility that, as with the price, wicks are represented at its extremes. If we observe a delta with a wick in its lower part, what it means is that at a certain moment of its development the difference in favor of the BID was extraordinarily large, and that at a certain point it has begun to negotiate much more actively in the ASK column, generating that reversal that leaves its mark in the form of a wick.

As it happens in general terms with the analysis of the footprint, although its interpretation is not as basic as I have explained above, it can provide us with certain usefulness at certain specific moments.

4.5 THE ORDER FLOW PROBLEM

As a basis, we must be clear about a series of ideas that will help us to understand the rest of the content:

1. A purchase is matched to a sale and vice versa.
2. An aggressive order is matched with a passive order.
3. In the columns of the BID and the ASK only the aggressiveness is reflected.

Considering the different types of orders, it is now necessary to understand what mechanism is used to match orders between participants and in which column this execution is reflected.

Order type	Matches with	Appears in
Buy Market	Sell Limit	ASK
Buy Limit	Sell Market	BID
Buy Stop	Sell Limit	ASK
Buy Stop Limit	Sell Market	BID
Sell Market	Buy Limit	BID
Sell Limit	Buy Market	ASK
Sell Stop	Buy Limit	BID
Sell Stop Limit	Buy Market	ASK

- ❖ Stop orders become Market orders when they are filled.
- ❖ Stop Limit orders become Limit orders when the price set by the Stop order is reached.

When the trader executes a Buy Market order, the order processing mechanism kicks in and goes to the Order Book to find the first Sell Limit order located in the ASK column to match this buy.

The same happens when a Sell Market order is executed. The processing mechanism directs that order to the most immediate price level of the BID column to find the counterpart in the Buy Limit orders that are pending to be matched there.

With limit orders the process is the same. A participant leaves his order pending execution in one of the two columns and it will stay there until an aggressive trader arrives who needs to match his order.

This is essentially what happens over and over again at high speed. Regardless of the type of order used to enter the market, the end result will always be that an aggressive order will cross with a passive one:

Buy Market	X	Sell Limit	=	ASK
Sell Market	X	Buy Limit	=	BID

And the column on which those order crossovers will be displayed will depend on the order that initiated it. Therefore:

- ❖ Buy Market crosses with Sell Limit and is displayed in the ASK column, since the order that initiated the transaction is the aggressive buy.
- ❖ Sell Market intersects with Buy Limit and is shown in the BID column since the initiation comes from the aggressive sell side.

Now let's do a reasoning exercise using as an example a trader who enters the market with a short position (sell). This trader has different ways to exit this position:



- ❖ Through a manual exit, either in loss or in profit by executing a buy Market order (and it would appear in the ASK).
- ❖ By executing the stop loss (stop loss), whose order will be a buy stop (and would appear in the ASK).
- ❖ By executing the take profit, whose order is a buy limit (and would appear in the BID).

Similarly, a trader who enters the market with a long position (buy) can leave it through three options:



- ❖ Through a manual exit, either at a loss or at a profit by executing a sell market order (and would appear in the BID).
- ❖ By executing the stop loss, whose order will be a sell stop (and would appear in the BID).
- ❖ By executing the take profit, whose order will be a sell limit (and would appear in the ASK).

The point of this example is that the same action, such as closing a position, can be shown in different columns (BID and ASK) depending on the type of order used for it.

Understanding this information is of tremendous importance because many analyses of order crossing are erroneous because they start from wrong premises.

The first conclusion should therefore be that not everything that appears executed in the ASK column is buying with the intention of

adding buying pressure to the market and not everything that appears in the BID column is selling with the intention of adding selling pressure. Herein lies the problem when analyzing order flow in any of its variants.

These programs based on the order crossing mechanism are set up to always reflect aggressiveness, the problem is that it is not possible to distinguish what intention is behind the executed orders.

When we see a cross executed in the ASK it will always be a Buy Market with Sell Limit order; while when we see a cross executed in the BID it will be a Sell Market with Buy Limit order, but what we will not know is the origin/intention behind such a cross order:

Intent of the orders executed at the BID	Intent of the orders executed at the ASK
Aggressive selling	Aggressive buying
Passive buying	Passive selling
Manual closing of a buy	Manual closing of a sell
Stop Loss of a buy	Stop Loss of a sell
Take profit of a sell position	Take profit of a buy position

The main source of error when dealing with order flow analysis comes because there is the belief that everything executed in the ASK has a buy initiative origin and everything executed in the BID has a sell initiative origin, but as we have just seen, nothing could be further from the truth. This type of software makes a reduction to the execution of aggressive orders with passive, but they cannot know what the origin/intention of those orders is.

What would happen if there is a crossing of orders of a Stop Loss

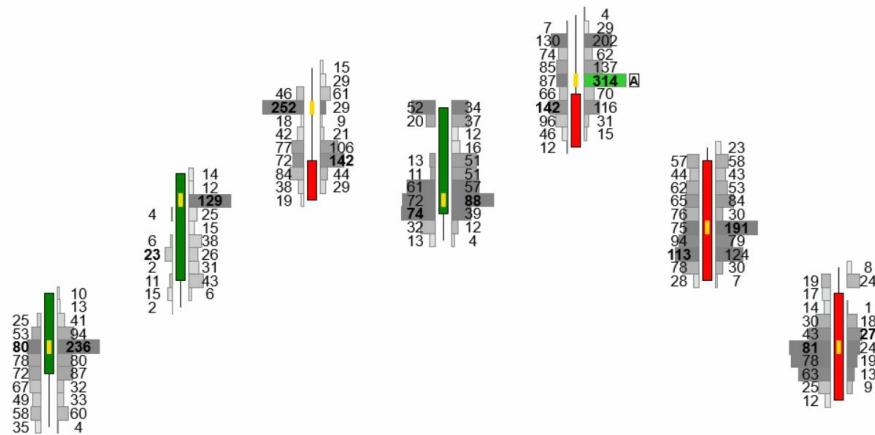
of a Buy Market and a Take Profit of a Sell Limit? This type of crossover will be reflected in the ASK, but is there really an intention to add buying pressure to the market? Obviously not, as we see in this example both traders would be out of the market and yet their transaction would be reflected in the ASK. This is the problem with Order Flow: it remains a highly subjective tool. Even more so when you do not know perfectly how order crossing works..

The same problem is encountered for the BID column. There could be the possibility of a matching of orders from someone who has triggered the stop loss of a buy position (Sell Market) with someone who wants to take profit from a sell position (Buy Limit). This crossover would be reflected in the BID but both are out, there is no new selling pressure.

I will now present two different contexts to exemplify again the Order Flow problem:

4.5.1 PROBLEM #1 PRICE DIVERGENCE

For example, if we analyze the chart of the footprint we see a bullish development in which at the top there is an imbalance (green bottom) in favor of the ASK with a downward turn immediately after; this fact offers us different interpretations.



Some will say that they are trapped buyers (assuming this imbalance in the ASK as aggressive buying with directional intent); others will say that they are executions of stop loss orders of long positions; still others will say that they are profit taking of long positions; and finally some others may say that it is a passive entry of sellers (takeover through sell limit orders).

All of them are probably right. And the truth is that most probably there is a bit of all of them. Also, at that point most likely the delta is negative, showing us a divergence.

This is where the problem becomes apparent when analyzing Order Flow. The truth is that in real time we cannot determine exactly what the true origin of these executions is. In many cases, in order to justify a proposed scenario, one of these particular reasons will be alluded to. For example, someone who is looking for a bearish move or who is already positioned short will see those large orders executed on the ASK and assume them to be "trapped buyers", as that is the reason that would justify their bearish approach.

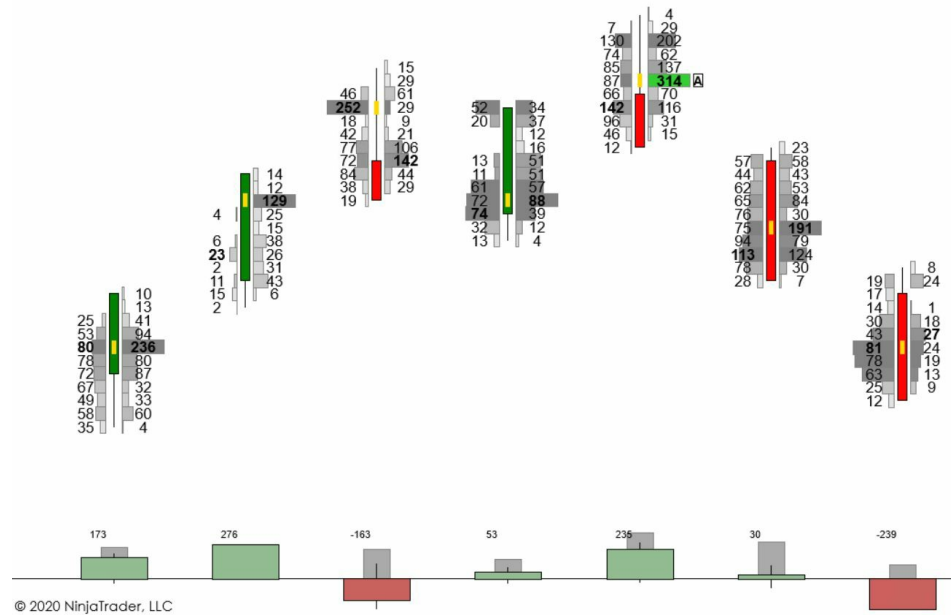
The only objective in this example is that, as it appears in the ASK column, it is a cross between Buy Market and Sell Limit orders; but to

state categorically that it is any of the possible origins already described does not seem to be a very solid approach.

Hence the importance that, in the event of deciding to work with Order Flow, the most logical thing to do would be to subordinate its analysis to the context provided by another approach such as, in our case, the Wyckoff methodology. The reason for this is that due to the complexity and nature of order matching we are going to find this type of imbalance in any part of the chart and this does not offer us an advantage.

4.5.2 PROBLEM #2 DELTA DIVERGENCE

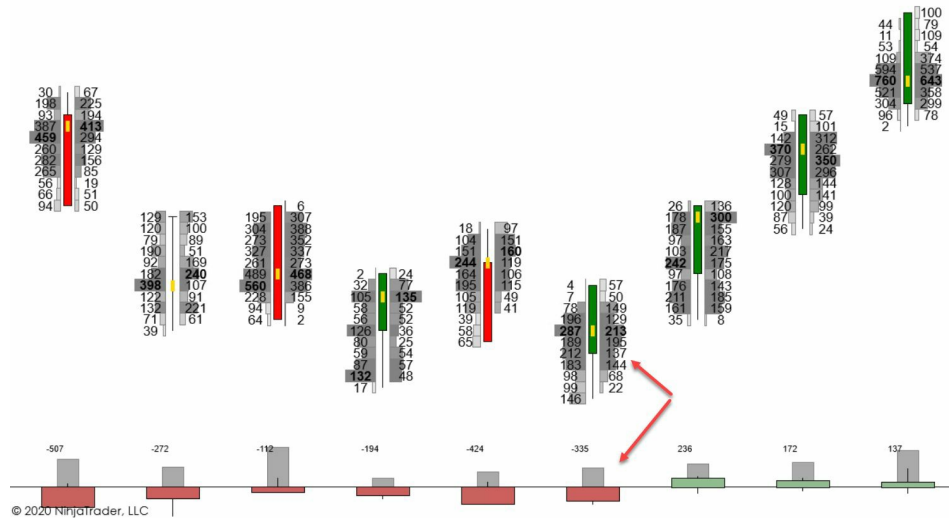
What happens when the Delta is not in agreement with the price? In a positive Delta its candlestick is expected to be bullish; and in a negative Delta, bearish. Divergence would appear when we observe a negative Delta on a bullish candlestick or a positive Delta on a bearish candlestick.



If all that appears on the ASK were market buying with the intention of adding upward pressure, it is impossible for a positive Delta to result in a bearish candlestick.

Continuing with the same example as above, we see that the bearish candlestick that originates the turn has a very positive delta (+235).

The reasoning behind this situation could be as follows: The positive delta may be the result of many aggressive buys (Buy Market) that have been blocked with passive sells (Sell Limit) and have not allowed the price to rise. All these crossing orders appear in the ASK. Subsequently, if there is little demand in the BID (few Buy Limits), a few aggressive sales would cause the price to move downwards. And this is one way in which a positive delta with a bearish candle would eventually be observed.



As you have probably concluded, delta divergences implicitly identify a takeover so if they appear in the right place they usually anticipate interesting turns. This does not mean by any means that all divergences will establish turns since sometimes these will occur over an area of little interest and without that intention of takeover behind, hence the problem of its use arbitrarily.

4.5.3 PRICE AND VOLUME TRADER

In the end, our task as traders is to identify when imbalances between supply and demand occur and these will eventually show up on the price and volume chart.

A trader who only takes into account the price and volume action may enter the market with some delay and may not have certain information available (the real interaction between the participants) but his trading will be much calmer as he will not have to interpret those order

crosses.

In the previous example of price divergence, the trader who simply analyzes the price action and volume will only focus on the fact that an anomaly has occurred in that action, a divergence effort result. That large number of executed orders will most likely be accompanied by an increase in volume and a narrow range already denoting some divergence. In addition, a subsequent downward turn would confirm this anomaly.

Beyond assessing whether there has been a jump of stops, profit taking, entry of short positions or that buyers have been trapped, which as we have already mentioned is probably a bit of all of the above, what is relevant is the final action and the trader who does not observe the flow of orders but knows how to interpret the chart will eventually reach the same conclusion, albeit with less stress.

4.5.4 CONCLUSION

In addition to everything discussed with respect to order matching, it is the precise moment to also recall the different types of agents that operate in the market and the intention behind their actions (hedging, speculation and arbitrage). The orders executed by these participants are also displayed on the BID and ASK, and as we have seen, not all of them have directional intentions which are ultimately those seeking price movement.

This is no small matter as the only ones who will show up again to defend their position in case they have aggressively entered the market

seeking a profit from the price move will be the speculative traders. We may see the execution of a large order at a price level and it may come from an institution with the objective of covering a position held in another parallel market, or it may be the activation of an arbitrage strategy, to name a few possibilities.

We therefore add a new layer of opacity and subjectivity. We have on the one hand that not all participants come to the market with a speculative interest; and on the other hand that the matching of orders cannot determine the origin of such trading.

We come to the conclusion therefore that the use of Order Flow independently could be totally meaningless since in no case can it offer us what is the most important aspect to determine in the market: the context; knowing exactly where we are going to look for trades and in which direction. Trying to understand this is vital in order to be able to perform solid analysis and scenario planning.

PART 5. VOLUME PROFILE

The Volume Profile is a variant of the Market Profile®, a tool designed by J. Peter Steidlmayer in 1985 for the Chicago Board of Trade (CBOT®). Steidlmayer was a trader and executive member of this important futures and options market for over 40 years. This new method of auction representation was initially intended only for CBOT members although it quickly spread outside. We can therefore intuit that his approach to how the market moves does not appear to have been ill-founded.

Unlike the Order Flow analysis, the Volume Profile is totally objective as it does not require any interpretation and therefore provides us with very useful information for our analysis and scenario approaches.

With the Volume Profile analysis we come back to all the concepts initially presented in the Auction Market Theory. We do not focus on

determining the intent of a particular order crossing, but rather broaden the picture to identify the most relevant trading zones.

The Volume Profile is not an indicator. It is simply another way of representing volume data. It identifies very clearly and precisely the number of contracts traded at different price levels.

5.1 AUCTION MARKET THEORY + VOLUME PROFILE

The Volume Profile uses the principles of auction theory to put it into practice and to visualize the areas of interest on the chart. Interest is simply measured by the activity that has been generated in a particular zone; and this activity is identified by the volume traded.

This tool will therefore help us to identify the zones of greater and lesser interest and will help us to evaluate the price when interacting with them in order to determine whether acceptance or rejection is taking place.

All these principles are based on the premise that the market has a memory and tends to repeat behavior. It is therefore expected that in the future certain areas will again behave in the same way as they did in the past.

One caution to keep in mind is that the market's memory is primarily short term. This means that more recent trading zones are more important than older ones. If the price initiates an imbalance the first zone to be taken into account will be the most immediate previous balance zone.

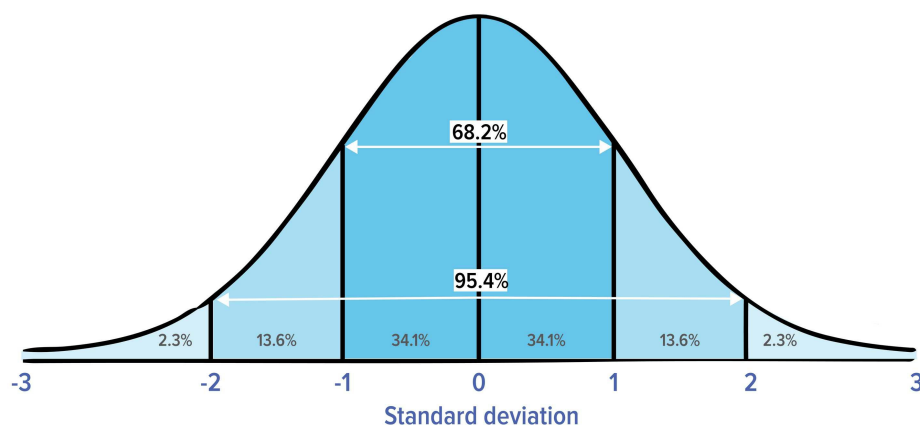
The longer the price has been away from a certain acceptance area, the less significant it will be. If we have no other reference, it will still be useful to value it, but it is important to be aware that the most immediate equilibrium zones will most likely be the ones that the market will look for in the first instance since they are the ones that best represent the value at the present moment.

5.2 VOLUME PROFILE COMPOSITION

Volume profiles are visually observed on the chart as a horizontal histogram where their values are distributed according to the trading at each price level.

Depending on the number of contracts traded at each price level, the shape of the distribution will vary. The more trades, the longer the length of the horizontal line; while a short horizontal line represents little trading.

As a reference we will take a normal or Gaussian Bell distribution to understand the most important statistical concepts:

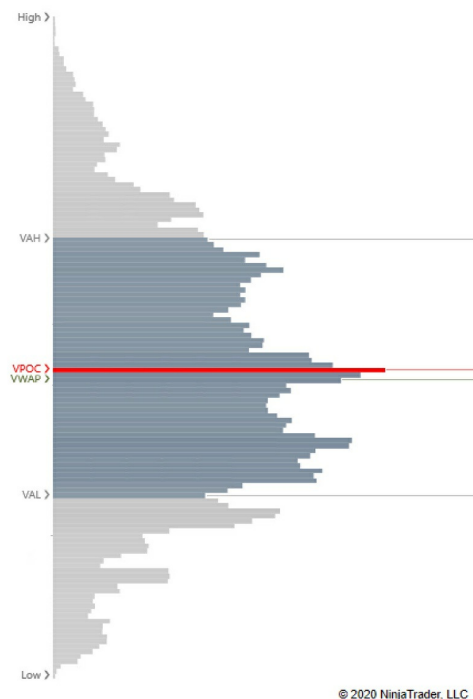


- ❖ The data are symmetrically distributed with respect to the

central point where mean, median and mode coincide.

- ❖ It has three standard deviations on each side, which are equally spaced and measure the amount of variability or dispersion around an average. It is also a measure of volatility.
- ❖ The first standard deviation comprises 68.2% of the data and up to the second deviation it reaches 95.4%.

VALUE AREA (VA)



This real example is in the form of a normal distribution where the values are distributed up and down around the central point.

The data are organized by a vertical axis on which the **price** is located and a horizontal axis representing the **volume** variable.

The value area is determined between the **Value Area High** (VAH) and the **Value Area Low** (VAL), is part of the first standard deviation and represents exactly 68.2% of the total volume traded in that profile. It is the most traded area of the profile and is therefore considered the area of acceptance.

The volume traded outside the value area comprises the remaining 31.8%. This is the least traded area of the profile and is therefore considered a rejection area.

The high and low levels of the value area (VAH and VAL) will act as support and resistance areas as some interaction is expected above them.

The breadth of the value area leaves clues about market conditions. A wide Value Area suggests that there is a large participation of all traders, everyone is buying and selling at the prices they want; while a narrow Value Area is a sign of low activity.

EXTREMES

This is the highest (High) and lowest (Low) price reached in that profile. These price levels should always be viewed as key reference points.

Depending on the trading generated at these extremes we can consider that they represent finished or unfinished auctions.

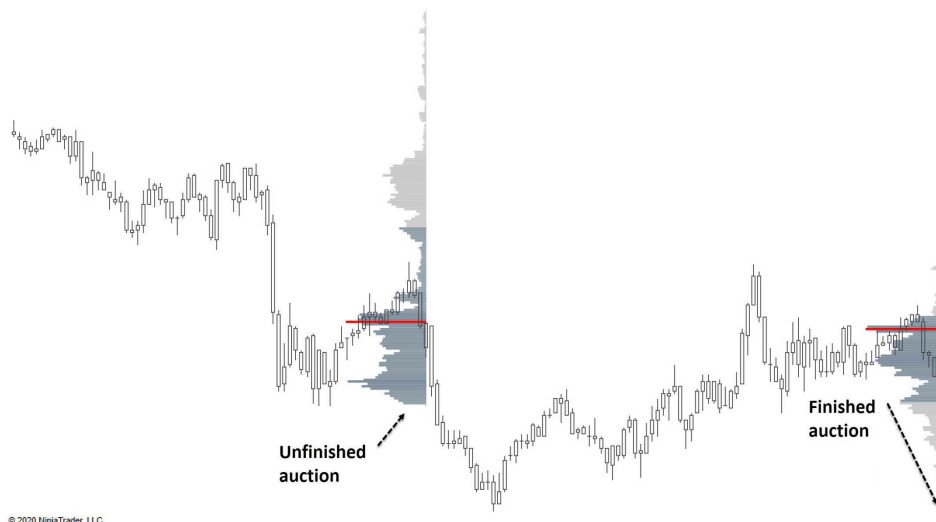
Finished auction. It is visually observed with a decreasing negotiation towards the end. It represents a lack of interest as the price

reaches price levels further away from the value zone, finally suggesting a clear rejection of the market to trade in that zone. By its very nature it is a Low Volume Node.

Prices have reached a point where some traders have considered it as an advantageous opportunity and have entered causing this rejection. The lack of participation from the opposite side is represented by this decrease in volume.

UNFINISHED AUCTION

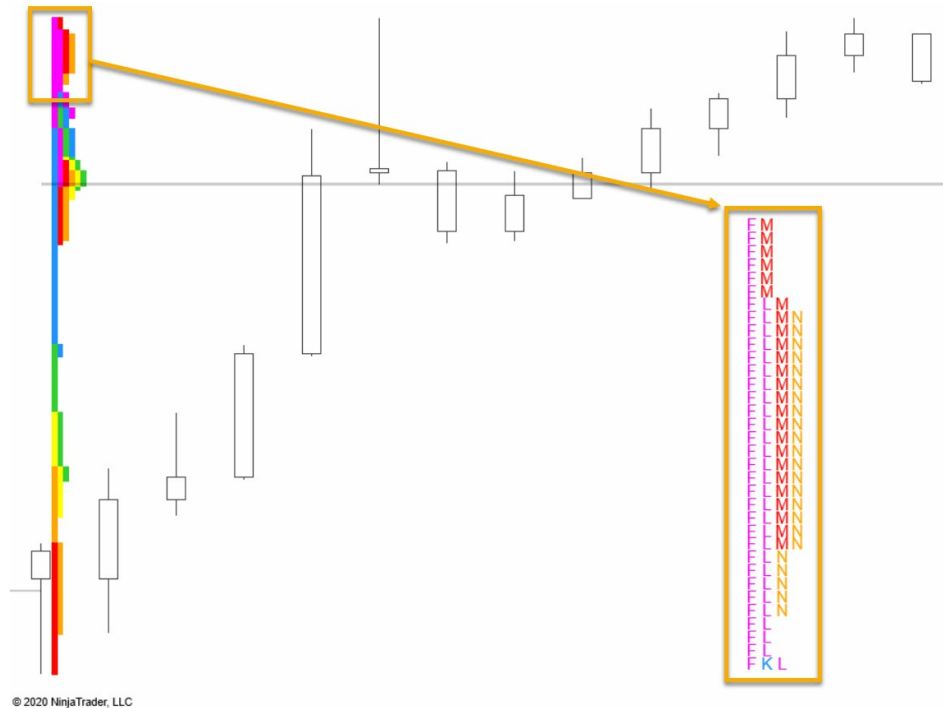
It appears as a High Volume Node on the end of the profile. Implicitly it represents interest to negotiate in this zone and therefore suggests a later visit of the price in case it has previously moved away from it. In the future visit it will be necessary to evaluate the intention behind it since it could develop with the objective of finishing the auction and turning around, or with the objective of continuing trading and continuing in that direction.



This concept of finished or unfinished auction can be very useful because if, for example, we are assessing the possibility of the price leaving an equilibrium zone upwards, we will want to see that at the bottom of that zone there is a finished auction that would suggest a lack of interest in trading there. In case of observing a possible unfinished auction it would be convenient to quarantine the scenario since it is most likely that before starting the upward movement there will be a visit of the price in that low part with the objective of testing the interest in that zone.



When in doubt as to whether we are in a possible finished or unfinished auction, it is advisable to treat it as finished. Unfinished auctions should be very visual and should not be involved in much subjectivity. Generally we are going to observe them as an abnormal cut in the distribution of the profile and in many occasions it will coincide with one of the two limits of Value Area.



In Market Profile this concept is totally objective: an unfinished auction (Poor High & Poor Low) appears at one end where at least two TPOs are observed; in other words, a finished auction will be represented with a single TPO at the price level (Single Print).

VOLUME POINT OF CONTROL (VPOC)

This is the level of highest volume concentration in that profile. It represents the price most accepted by both buyers and sellers (fairer) and sets the level from which the value area is calculated.

Since most of the volume comes from large institutions, this is where these large traders have accumulated most of their positions. They generally accumulate contracts in a range of prices but the VPOC represents a benchmark as it identifies where the most interest is.

As it is a level that will attract a lot of trading, it is generally advisable to avoid trading in its vicinity. The broad consensus among participants will cause fluctuations around this level. This behavior will continue until new information appears that unbalances the participants' perception.

The VPOC allows us to establish who is in control of the market. If the price is above it, we will determine that the buyers will be in control, so it would make more sense to trade long; if it is below it, the sellers will be in control, so trading short would be a better option.

Keep in mind that the VPOC, by its very nature, will always be a High Volume Node, but not all High Volume Nodes will be VPOCs.

VOLUME WEIGHTED AVERAGE PRICE (VWAP)

If there is one level widely used by large institutions it is VWAP. Huge transactions seek to execute at the price level where the VWAP is found and that is why it has elevated its level of importance.

The VWAP represents the average price of all contracts traded during a particular time period. The formula for obtaining it is as follows:

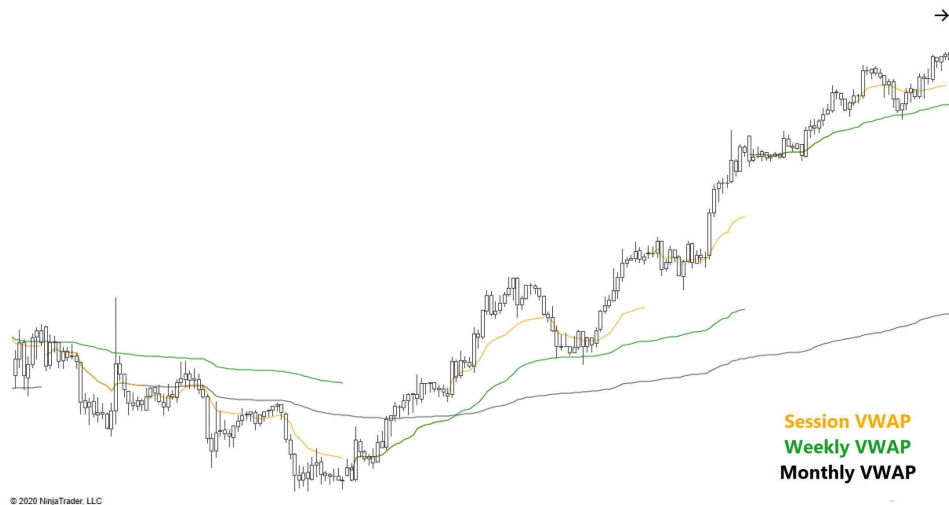
$$\text{❖ No. of contracts traded} * \text{price of the asset} / \text{Total contracts traded}$$

To understand it a little better, we can say that above the VWAP there is the same volume traded as below it, i.e. it represents an important equilibrium level. This equilibrium means that when the price reaches the VWAP there is the same probability that the price will go up or down.

It is displayed on a chart like a traditional moving average and its position varies as trades are executed. Generally, depending on the trading style, the VWAP of the session, the weekly or the monthly is used.

The VWAP is used by institutional traders primarily as an average to determine the value of the asset at that point in time so they consider that they have bought low if the price is below and high if it is above.

Institutions have taken the VWAP as a reference measure with which to judge the quality of their executions, hence its relevance and the fact that we treat it as an important trading level.



When they receive an order, they do not execute all the contracts they need at one time, but will try to do so gradually, knowing that their work will be judged on the basis of this reference level.

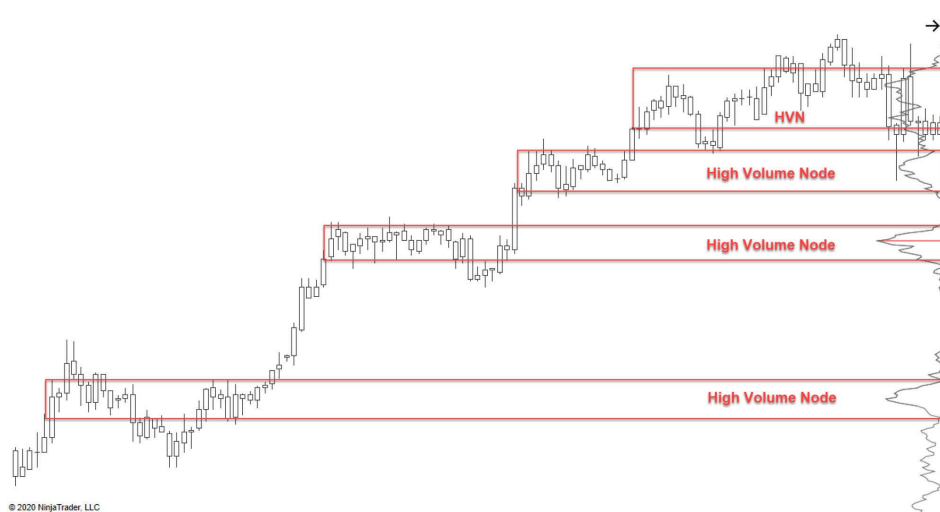
Since it represents an important equilibrium level or fair price, it is a good measure of whether we are buying too high or selling too low. We can tell this by adding one or two standard deviations to the mean. Just because the price is at some standard deviation does not mean that it cannot continue to move in that direction, we could simply use it as one more footprint to add to our analysis.

But be very careful because everything is subject to the valuation of the market at that moment. In a market in balance a price below the VWAP will be considered cheap and a price above it expensive; but just as the market becomes unbalanced to one side or the other the VWAP no longer represents efficiency as now the perception of value has changed.

Depending on the time frame we can make use of different levels of VWAP. The most commonly used are the session VWAP for intraday traders and the weekly and monthly VWAP for medium and long term traders.

HIGH VOLUME NODES (HVN)

High volume nodes. These are zones that represent balance and high level of interest by all market participants as both buyers and sellers have been comfortable trading there. It is observed as peaks in the volume profile.



Although we have used a composite profile for this example, the

fundamentals are equally valid and applicable to all profiles.

Past balance zones act as magnets attracting the price and keeping it there. As in the past there was some consensus between buyers and sellers, in the future exactly the same is expected to happen. That is why they are very interesting areas for target setting.

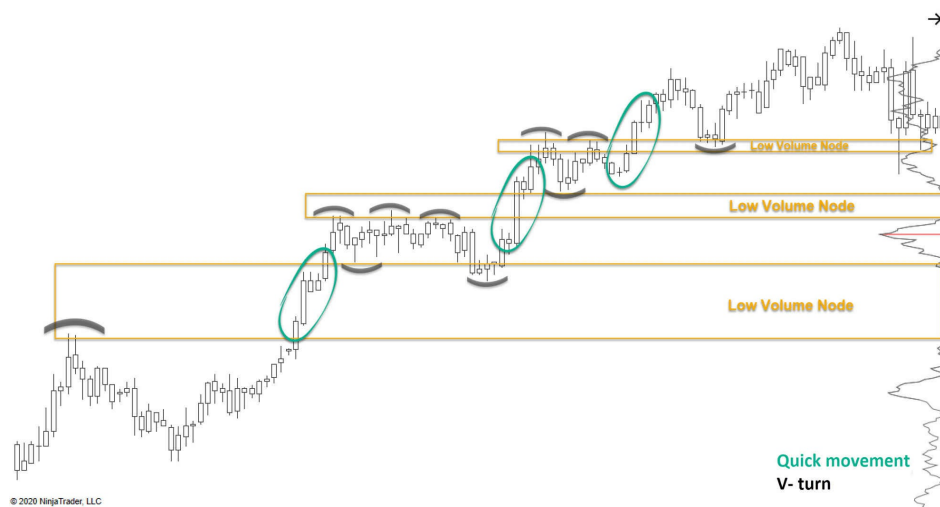
Different High Volume Nodes can be identified within the same profile.

Low VOLUME NODES (LVN)

Low Volume Nodes. These are areas that represent imbalance/rejection. Neither buyers nor sellers have been comfortable trading and therefore are considered in some way "unfair" prices. They are observed as valleys in the volume profile.

As in the past there was no consensus, it is expected that in the future there will be no consensus either and cause some rejection, so they are interesting areas of support and resistance where to look for potential entries.

It is important to understand that rejection can be represented by the price in two ways:



V-TURN

The perception of the value has not changed with respect to the previous equilibrium zone and there is a rejection to trade at these levels. The market turns completely around to re-enter the previous zone in which buyers and sellers feel comfortable trading.

What triggers this price reaction is firstly the placement of passive orders waiting above that area to block the move, coupled with a subsequent aggression that confirms the V-turn and return to the previous value area.

Visually it may be observed in the price as prominent wicks at the ends of the candlesticks which will suggest such rejection.

RAPID MOVEMENT

Participants' perception of value has changed and is represented in the price by a violent movement. The market, based on the new information, rejects trading at these LVN levels and moves quickly through them.

Technically, what causes this rapid movement is, on the one hand, the execution of stop loss orders by those who are positioned on the opposite side; and on the other hand, the activation of momentum strategies that enter aggressively with market orders.

Visually, it will be observed on the chart with wide range candlesticks generally accompanied with high volume.

As with HVNs, more than one Low Volume Node can be displayed within the same profile.

5.3 TYPES OF PROFILES

The volume profile is a tool that can be adapted according to the operator's needs.

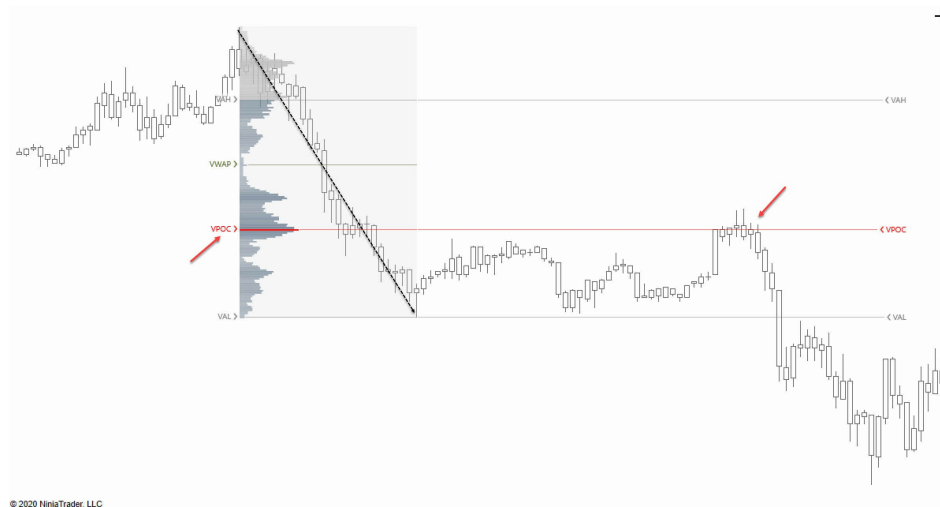
The main difference when using one type of profile or another will be determined by the trader's timeframe and the context he needs to cover in his analysis.

Basically we can differentiate three types of profiles:

FIXED RANGE

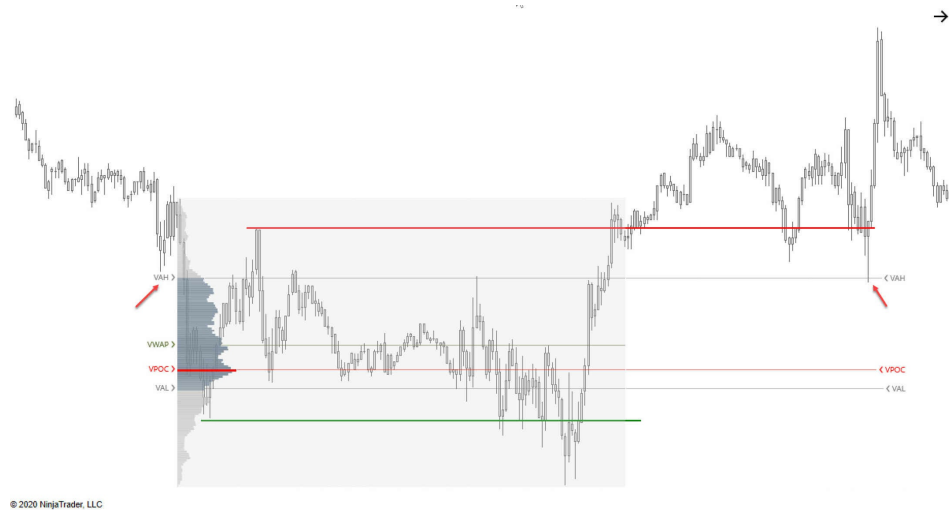
This type is very versatile. Its peculiarity is that it allows us to launch profiles manually on any particular price action.

It is especially useful for identifying trading zones in two types of contexts: trend movements and ranges.



If we see a bearish movement we can launch a profile of the whole momentum to identify the interesting zones over which the price is likely to make some kind of bullish pullback. It is in those trading zones where we want to be prepared to evaluate the possibility of entering in favor of the trend. In the example we see how the price makes a test to the VPOC of the bearish momentum and from there it generates a turn that causes the development of a new bearish movement.

If what we have is a context of lateralization, a structure of those we work under the Wyckoff methodology approach, the profile will come in handy to identify mainly where is the VPOC that determines the market control and the value zone with its extremes (VAH and VAL). These will be very interesting to take into account in order to look for the test after the breakout of the structure, as in the following example.

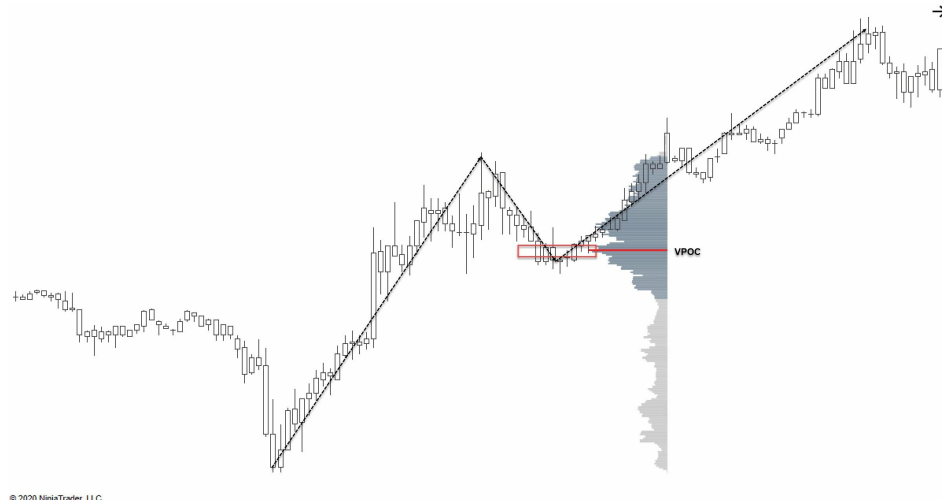


Unlike the other types of profiles, the fixed range is not updated and only analyzes the volume traded in the area to be determined.

Regardless of whether you are going to work with structures or not under the Wyckoff methodology approach, it is very useful to pull profiles that encompass several sessions in case we see an overlap between their value areas. If we see more than one session generating value over a certain price range, it is best to launch a profile that includes all the price action, since the operating levels that such a profile can provide us with will have by their very nature a greater relevance.

SESSION

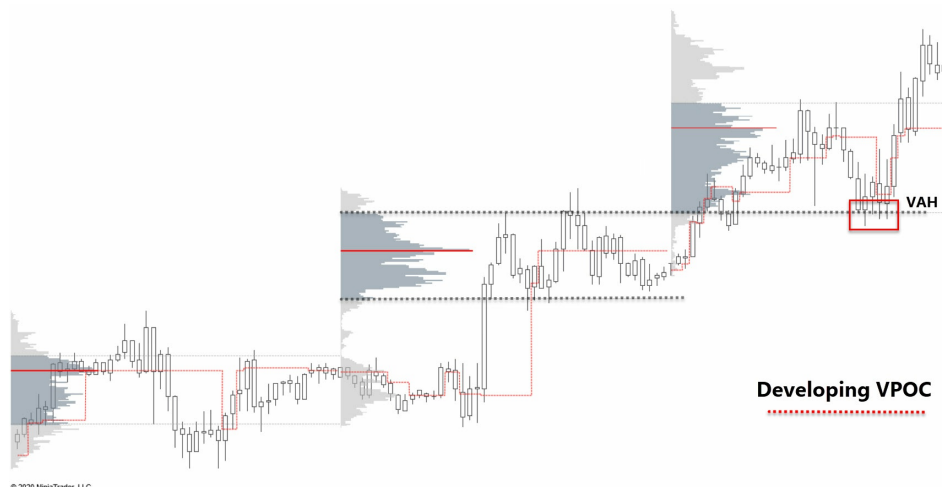
This is the profile of the day. It is especially useful for intraday trading where the most important trading zones of the session are taken into account. It ranges from the beginning to the end of the session and is updated as the day progresses.



© 2020 NinjaTrader, LLC

Shorter-term traders use both the levels of the session and those of previous sessions to make their scenarios.

If we are observing a bullish movement and a subsequent lateralization of the price, it would be interesting to look for the incorporation in favor of the trend movement on some operative level. As we know, the most important level of the whole profile is the VPOC so we must take it into account to wait for our entry trigger on it.



© 2020 NinjaTrader, LLC

Continuing with this same example, another interesting area on which to look for a test after a breakout could be an operating area of previous sessions. If we are above the value area of the previous session,

the first operating level on which to expect a test to look for the bullish continuation will be the Value Area High of the profile of the previous session.

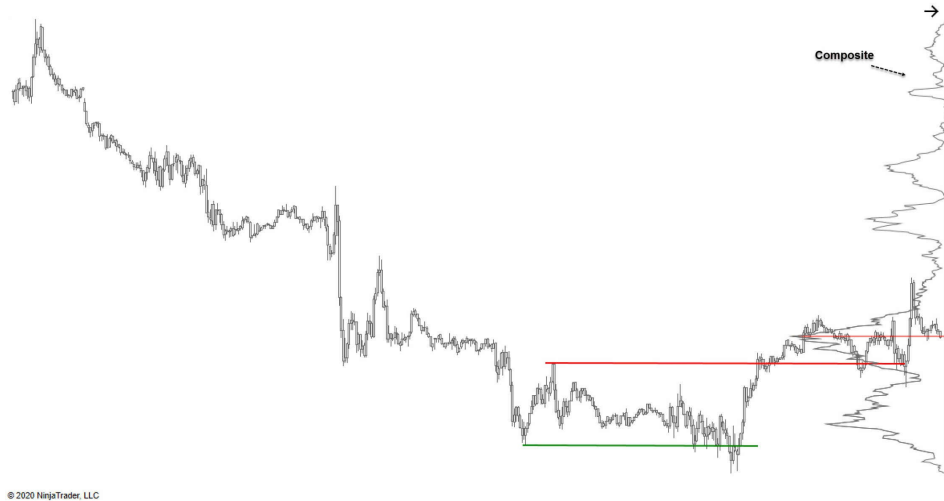
COMPOSITE

Originally the profiles were displayed only by sessions and this idea of grouping them was introduced by Donald L. Jones in his book "Value-Based Power Trading" in what he called "The Overlay Demand Curve". The objective was to try to eliminate the noise of the shorter term and thus get a better understanding of the market condition, of the context.

This type of profile can be set up in two ways:

- ❖ Fixed. Within the fixed mode we have the possibility to select the range of dates we want to include in the profile analysis. You may want to know the profile for the past week, the past month or the current year, this mode is designed for this particular requirement.
- ❖ Variable. The variable mode has the important peculiarity that it shows the traded volume of all price levels that are currently on the chart. It is important to keep this in mind because if the chart is moved, the profile will change.

The best use of this type of profile, regardless of the time frame in which we are trading, is to analyze the overall context and identify the trading zones (mainly the high and low volume nodes) that we have both above and below the current price.



These zones will serve to indicate the bias of the market in a more macro context as well as for the establishment of possible zones where to look for entries and exits.

If for example we are working on a structure, it is very interesting to analyze the profile of the Composite to identify the longest term High Volume Nodes on which to establish targets for profit taking.

Another use could be to identify in the macro context a large Low Volume Node and to encourage the development of a shock on it. We could be working a potential cumulative structure. If when analyzing the context we identify an LVN relatively close to the structure, it would be interesting to take into account the possibility of the Spring that generates the imbalance of the structure.

5.4 DIFFERENCE BETWEEN VERTICAL AND HORIZONTAL VOLUME

Many who look with some suspicion at the Volume Profile allude that it is not necessary to incorporate this type of tool in order to be able to make solid analyses and approaches based on volume data.

This is absolutely true; it is obvious that there is absolutely nothing necessary. The problem is that they fail to understand what kind of information they can acquire through their evaluation.

The first thing that should be conveyed is that the Volume Profile was not developed as a substitute for the classic volume. They provide different information and are therefore completely complementary.

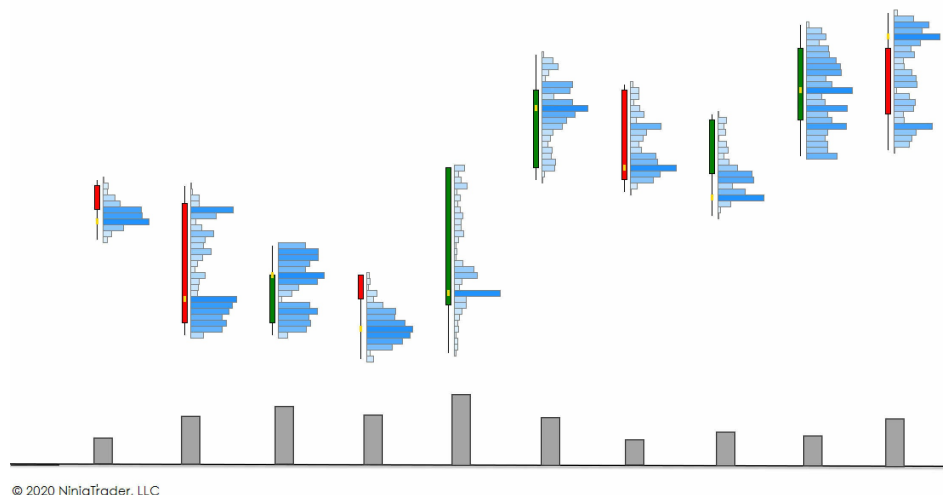
To really understand what information is provided by volume data, it is necessary to examine it from two points of view:

- ❖ **Volume in time:** This is the classic volume that is observed vertically on the chart. It has to do with the number of contracts exchanged within a certain period of time. It tells us when the major traders are active.
- ❖ **Volume in price:** This is the Volume Profile and is observed in

the form of horizontal bars. It indicates the number of contracts traded within a certain price level. It tells us where the activity of the major traders has taken place.

As we can see, both provide us with different information about the same action (professional activity), volume in time has to do with when, while volume in price has to do with where.

With the vertical volume we can know that during the development of a particular candle a specific number of contracts have been traded, but how has the trading been distributed through the different price levels? This is the information that horizontal volume gives us and that classical volume cannot. It is quite another matter if you need to know this information for your trading.



5.5 DIFFERENCE BETWEEN VOLUME PROFILE AND MARKET PROFILE

The main difference between the two tools is that the Market Profile is designed based on time, while the Volume Profile is designed based on volume.



The Market Profile represents the price data on the chart in letter format, where each letter (called TPO or Time Price Opportunity) is identified with 30 minutes of trading. So, the letter A will correspond to the first 30 minutes after the opening of the session, the letter B to the next 30 minutes and so on, adding letters and minutes until the end of the day.

Traders who base their trades on Market Profile analyze the opening of the day with respect to the value area of the previous day and the evolution of the Initial Balance (range that covers the first hour of trading) to determine the type of day that is likely to happen and propose scenarios based on this. It should be noted here that some traders determine the Initial Balance based on the first half hour only.

Although Volume Profile traders usually do not take the Initial Balance into account, the message it conveys can be very interesting mainly because:

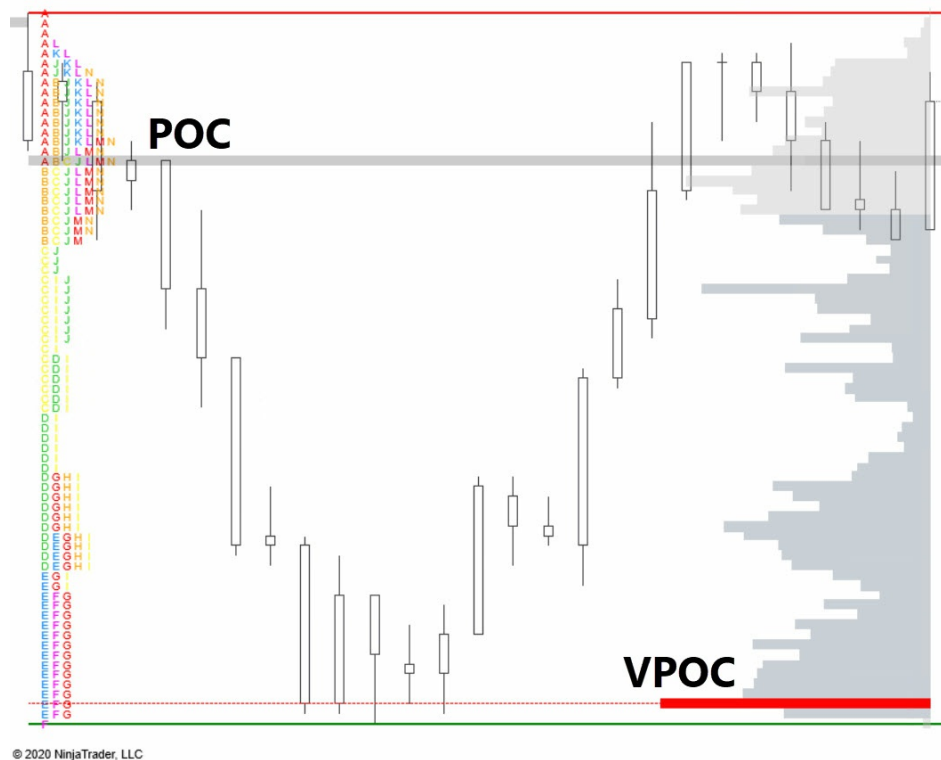
- ❖ The narrower its range, the greater the probability of a trending day; and the wider the range, the greater the probability of a sideways day.

An interesting use that MP offers us is the objective determination of acceptance or rejection on a price level. While in Volume Profile this can be shrouded in subjectivity, Market Profile analysis eliminates this discretionality: rejection is visualized by 1 TPO; while 2 or more TPOs begin to represent acceptance.

The development of the distribution of the profiles of both tools will tend to look quite similar if both tools will tend to look quite similar although they will not be exactly the same. This is obvious since they do not use the same data for their representation. An accumulation of TPOs

will indicate that the price has spent a long time at that level, while an accumulation of volume will tell us that a large number of contracts have been exchanged at that level.

In Volume Profile, as it is designed according to volume, the most traded level will not necessarily be the level where the most time has been spent; since the price may reach a level that in a few seconds accumulates a large number of orders and turns around (as in the example). The time that the price will have spent at that level is little but the volume traded is a lot so the Volume Profile POC will be at that level while the Market Profile POC will not.



Due to the current ecosystem of financial markets where the importance of volume is highlighted, it would seem more interesting to use the combination of price + volume (Volume Profile) rather than price + time (Market Profile).

This is not to say that the time variable is unimportant or less important, nothing could be further from the truth. Obviously, time is a key element in establishing where participants decide the value of a particular market is at a particular moment in time. The consumption of time in a given area is an unquestionable sign of acceptance and therefore value building.

5.6 PROFILES SHAPES

As is obvious, the market will not always develop D-shaped profile distributions as this would mean that we are in an infinite equilibrium context.

There is a lot of theory written in Market Profile about the different types of days based on the shape of the profile (normal day, normal variation, trend day, double-distribution trend day, non-trend day, neutral day, neutral day extreme).

The truth is that identifying the shape of such profiles to determine what kind of day we have had may be valid for the human mind from the point of view that we always want to control everything and need to look for a logic to every behavior; but from a trading point of view it does not seem to be a very useful approach since the categorization is done by an a posteriori analysis.

Furthermore, over time evidence has been provided that it is not possible to consistently predict what type of day we are most likely to have based solely on the previous day's categorization. Steidlmayer himself eventually acknowledged this. Unsurprisingly, it is impossible to know what shape the current session profile will have until the session has

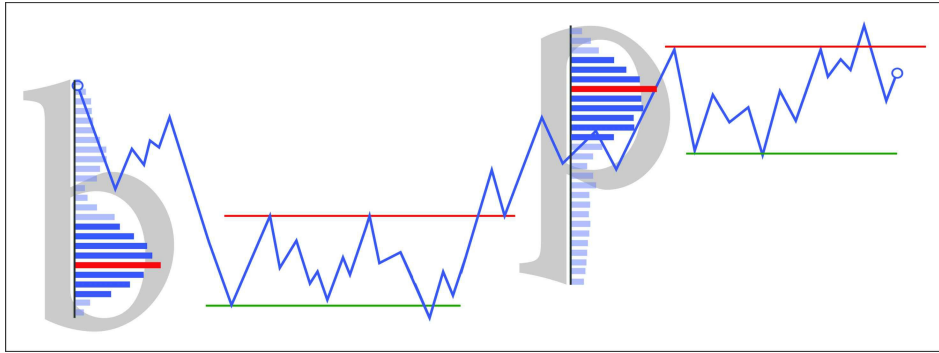
ended.

It is exactly the same with the labeling of events, phases and structures under the Wyckoff methodology approach. It may be useful for the more novices to feed the subconscious with the various ways in which the market can represent accumulations and distributions; but it is totally useless from an trading point of view as the confirmation of all this is done after the fact.

It would therefore seem more sensible, for the purpose of setting up trading scenarios, to focus on identifying the creation of a value zone (range) and to evaluate the price continuously as it interacts at its extremes in order to identify acceptance or rejection. The analytical tools offered by the Wyckoff methodology help us in determining who is more likely to be in control during the development of the range (buyers or sellers) and therefore in which direction the path of least resistance lies.

During the development of the moves, profiles will be generated and two very common ways in which trend and sideways behavior is graphically observed are the b and P patterns.

These patterns represent the first three phases of market activity that Steidlmayer presented in his first studies and that by Wyckoff methodology we identify as Phases A and B of the development of structures.



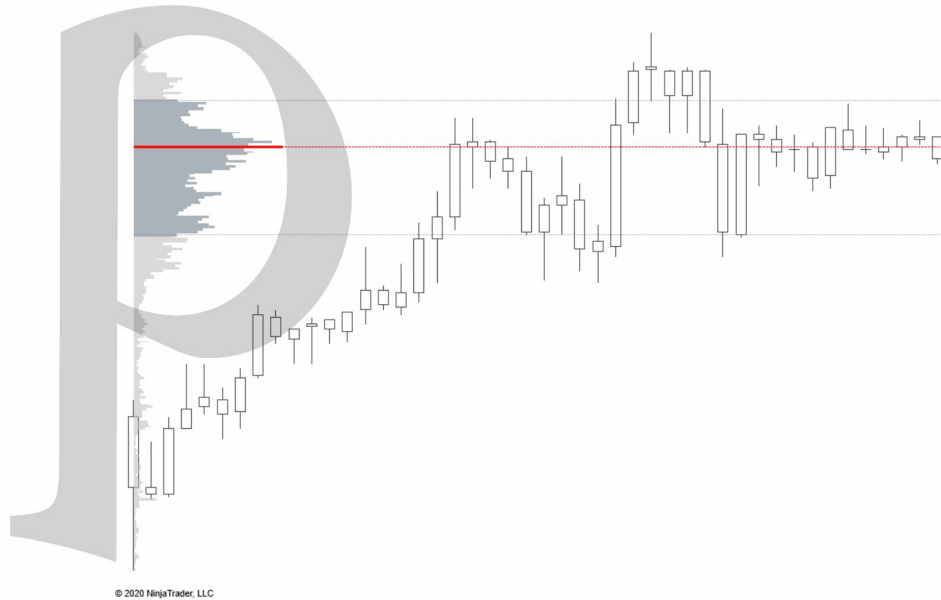
These patterns will alert us to the halt of the previous trend movement as well as the new lateralization context. These two types of profiles show the same behavior but in both directions. Firstly, in the low price zone the price moves with a certain fluidity developing the trend movement until it meets traders willing to trade in the opposite direction. At that point an equilibrium range begins to develop, a zone of high participation that generates the profile bell.

As long as the price is during the trend movement, we should only be looking to trade in favor of that direction. We can rely on the developing VPOC and the rest of the operating levels.

5.6.1 P-SHAPE PROFILE

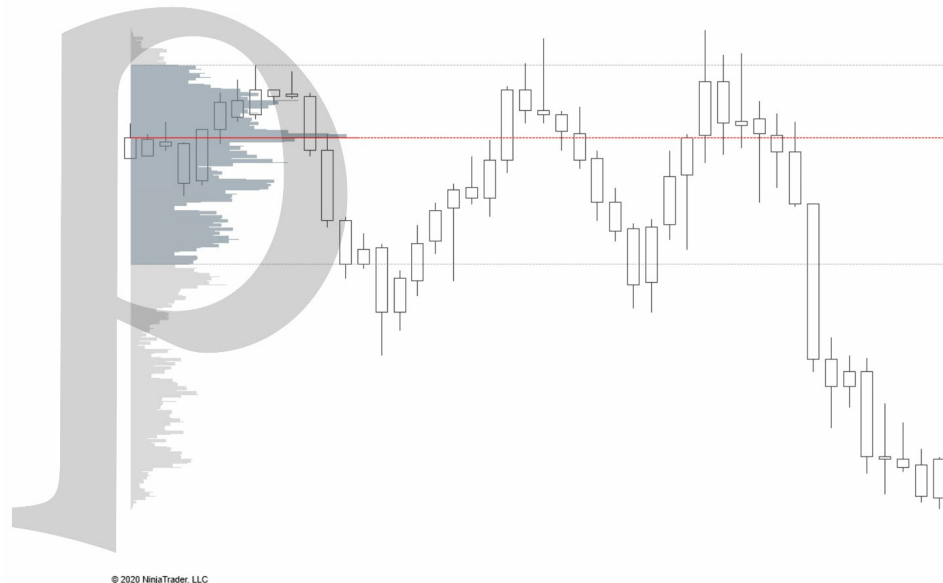
Characteristic of bullish movements and representative of future distributive and reaccumulation schemes.

This type of profile suggests strength on the part of buyers who have had the ability to push the price up with relative ease until it reaches a point where sellers begin to appear.



It is made up of two parts: a first one where the upward imbalance is observed and a second one where the market starts a rotation process (range). This is important to keep in mind because if such a process occurs in reverse (first a rotation and then a downtrend) visually we will still see a P-type profile with the big difference that it will have difficult trading validity.

All the concepts and tools that are being proposed have an operative sense; and in the case that concerns us with the P and b patterns they are only interesting if the last price action is the creation of new value (range) since it will be from there where the next profitable imbalance will be generated. That is, if we are at the end of the bearish movement at the point of having already identified the P pattern in mirror (with the imbalance to the right) completely, from the point of view of operating that imbalance we may be too late and the most likely thing is that the price will generate a new equilibrium zone as the most immediate action. On the contrary, if we have a theoretical P pattern (with the imbalance to the left) we will be able to take advantage of the subsequent trend movement (which may be upward or downward).



Besides looking at it in isolation (session profiles), it is interesting to know that a longer-term uptrend will be composed of several of these profiles in its internal development. In this case they will simply be reaccumulative patterns that give rise to the continuous upward development.

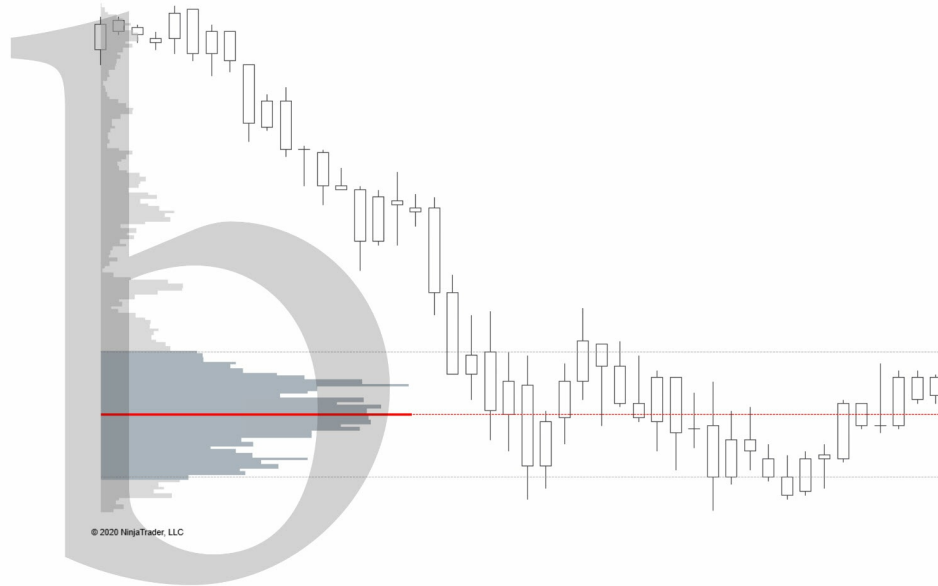
In case this type of profile appears after a prolonged downtrend, it could alert us to the imminent end of the downtrend at least temporarily. Here it is important again to note that it refers to see the theoretical P pattern; because if what we see is a mirror P pattern what we would actually have would be a distribution and therefore the downtrend would denote strength.

5.6.2 b-SHAPE PROFILE

Characteristic of bearish movements and representative of future

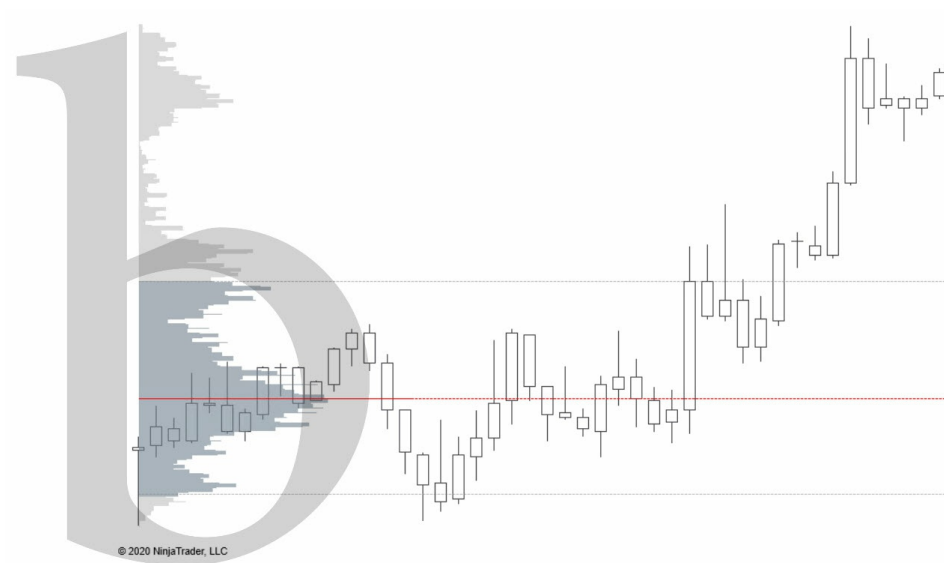
cumulative and redistribution schemes.

Individually, this type of profile signals an imbalance in favor of sellers. Sellers are in control and have pushed the price strongly downwards until finally some participants appear buying and a new rotation process is generated.



Within the context of a longer-term downtrend they will appear with redistributive character being excellent areas where to look for incorporation in the trend.

As with the P pattern, the theoretical type b is formed by a bearish imbalance as the first part and a value creation as the second. A rotation first and a subsequent bullish imbalance would also appear visually with a b-shaped profile but operationally we may not be able to take advantage of that first imbalance. On the other hand it would also not have the same implications when analyzing the health of the preceding uptrend, where what we are looking for is upside rejection and value generation below.



In case of observing a b-shaped profile after a prolonged bullish movement, it could signal the end of such movement and sometimes the beginning of a new one towards the bearish side.

5.7 USES OF VOLUME PROFILE

This tool offers us totally objective information that fits perfectly within the context provided by the principles of the Wyckoff methodology.

Some of the most important uses we can make are listed below.

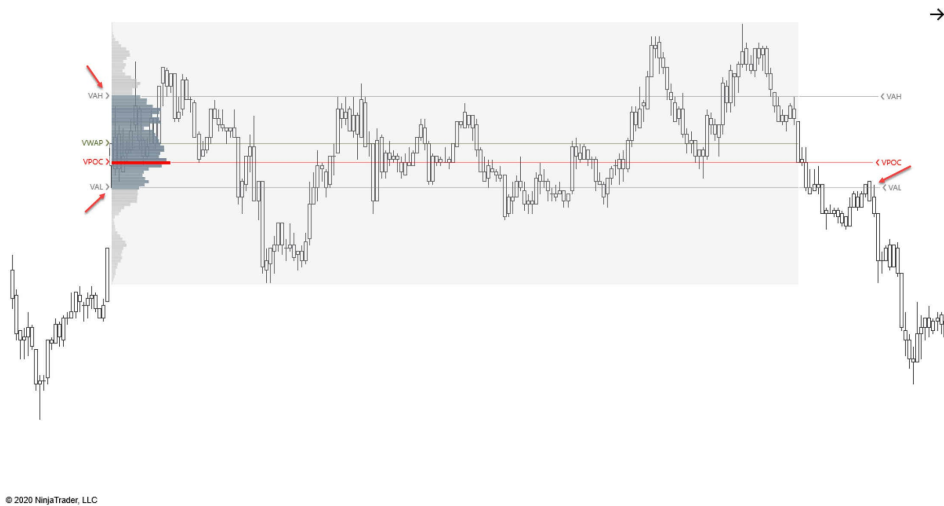
5.7.1 IDENTIFYING STRUCTURES

There will be occasions where the delineation of the ends of structures will not be very visual, possibly because the price action has generated unclear movements. In this context it can be very useful to use the Volume Profile to pinpoint the high and low zones of the value area (VAH and VAL) assuming this area as the range that is generating the cause of the subsequent movement.

By their very nature, the upper and lower limits of the structures we work with according to the Wyckoff methodology will always be low trading zones (LVN). The price turns that generate the creation of these supports and resistances are areas where the price has not wanted to trade

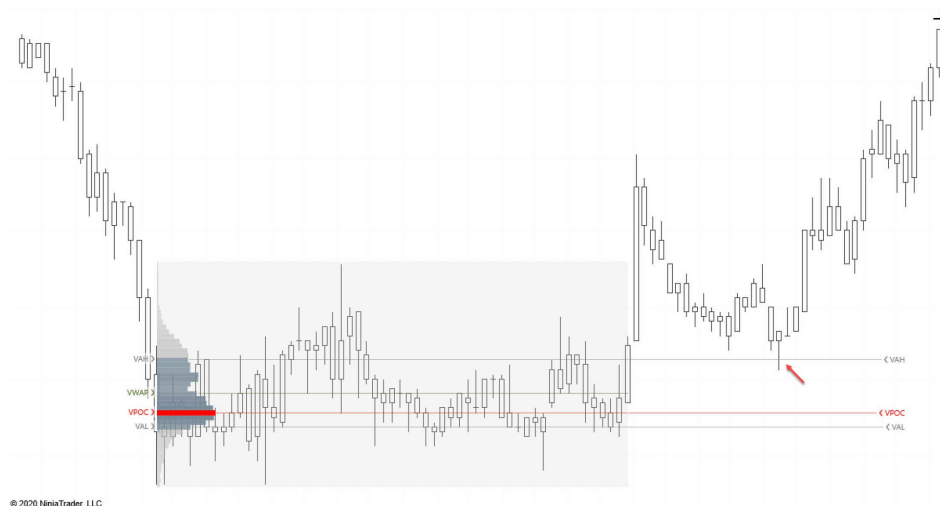
and are therefore identified as rejection. That rejection we already know is visualized within the volume profile as an LVN.

In addition, sometimes the natural ends of the structures (Creek and Ice) will coincide with the ends of the value area (VAH and VAL). By pulling a profile of that structure we will be able to observe how all price action contained within the range will be part of the value area.



As can be seen in the chart below, the stop of the bullish movement does not show very clear price action where the stop events are genuinely visualized. If we are in the final part of the development of such a structure we could pull a profile of the whole structure to identify key levels by Volume Profile. In this case we see how after the final shock from above the price quickly crosses the entire value zone including the VPOC and VWAP. At this point we should be favoring the distribution and therefore the first scenario in short would be to wait for the test after breakout (LPSY) on any of the operating levels. The first of these levels to take into account would be the Value Area Low as it is the first that the market would find. We see how it made this test to continue the bearish development from there. The next and last level to look for the potential LPSY would be the VPOC of the profile.

In this other example something similar happens. Stopping events may not be the most visual and then there is the added difficulty of framing that bullish movement that leaves the value zone and then re-enters. The objective is that, if we find ourselves seeing an imbalance after the development of a structure already with a certain maturity, we can launch a profile to identify the operating levels on which to expect the price to develop the test looking for the continuation of the movement. Again we see how after the breakout movement, the test is done just above the end of the value area of the profile, in this case above the Value Area High; it is the perfect area for the search of the long entry trigger.



The **scope of the profile** should include all the price action from the time the rotation starts until just before the imbalance occurs. There are traders who may also include the breakout action within the profile, which is not to say that this is incorrect. Just keep in mind that from the trading point of view what we are looking for is that this test after breakout is going to look for some trading level of the previous accumulation/distribution and this would leave out both the imbalance and the subsequent movement to the test.

As we can see, most of the structures that develop in real time

markets are not as genuine as shown in the ideal book examples, plus they are all different from each other. But this does not mean that they are not tradable. This is where the levels provided by the Volume Profile tool come into play.

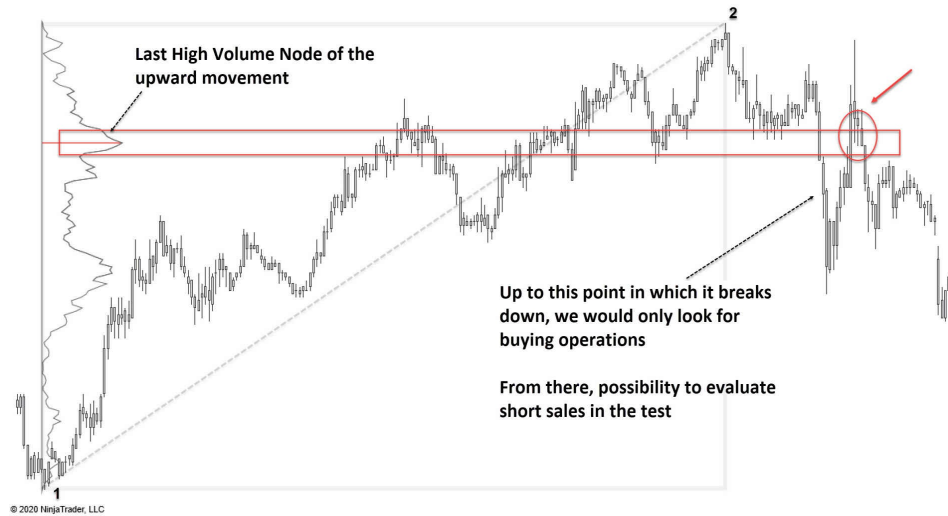
5.7.2 DETERMINING MARKET BIAS

BY ANALYZING THE TRADING ZONES

We will always favor trading in the direction of the last high trading node generated. And a scenario against this direction would only arise when the price has broken the zone that supported the last move.

If the price is above a high volume node (HVN) we will determine that the control is in the hands of the buyers and we will only raise a short scenario when the price crosses this zone from below, which will suggest that the control has changed in favor of the sellers.

The logic is that at these nodes the price returns to a state of equilibrium and we will not be able to determine in which direction it will move subsequently. Only after confirming the effective breakout of this zone would we be in a position to propose a scenario with some robustness.



The above chart shows this concept very clearly. If we are in the final part of the bullish movement, above the last HVN, we should wait for a breakout before looking for a short trade. The profile has been drawn covering only the bullish movement (from 1 to 2), since what we are interested in knowing is where is the volume node that is supporting this movement. When this is crossed downwards we can suggest that the control has shifted to be in favor of the bearish and now we are in a position to propose a short scenario for example to the test of this HVN.



In this other example we have the same dynamics but in reverse. We locate the HVN of the bearish movement in force and the idea is to

continue favoring shorts until it is broken. Note, not until this HVN is broken, but the last one to be generated. That is, if the trend continues downward, we must continue to update the profile to identify where the last HVN is and only at the time of its break to the upside could long entry be considered.

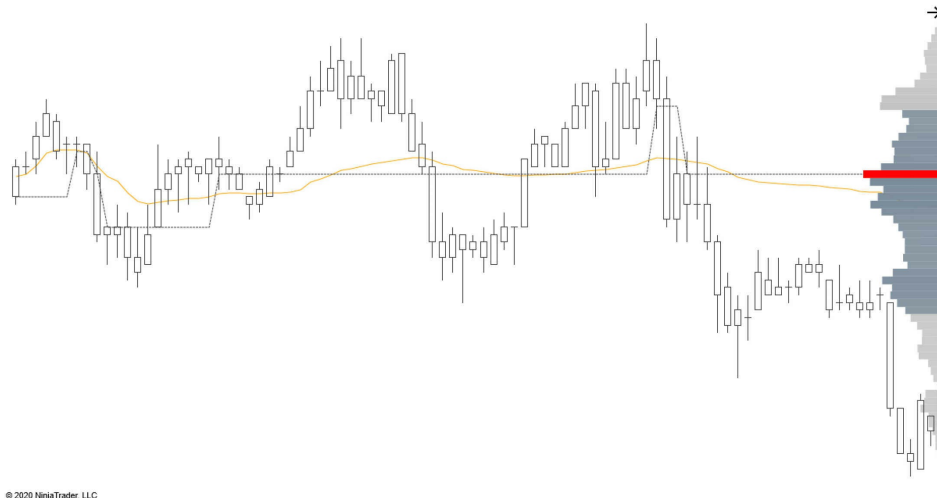
It should be noted that not necessarily the HVN that determines the control of the movement must also be the VPOC of the profile. We will simply take into account the last one to develop, regardless of whether it is the VPOC or not.

BY ANALYZING THE OPERATING LEVELS

As a general rule, the most advisable is to trade with as many operating levels in favor as possible. That is to say, if I am considering a long trading idea, I will want to have all the trading levels below the price and vice versa if I am considering a short idea. This context will suggest that the market is unbalanced in that direction and that it is therefore the path of least resistance.

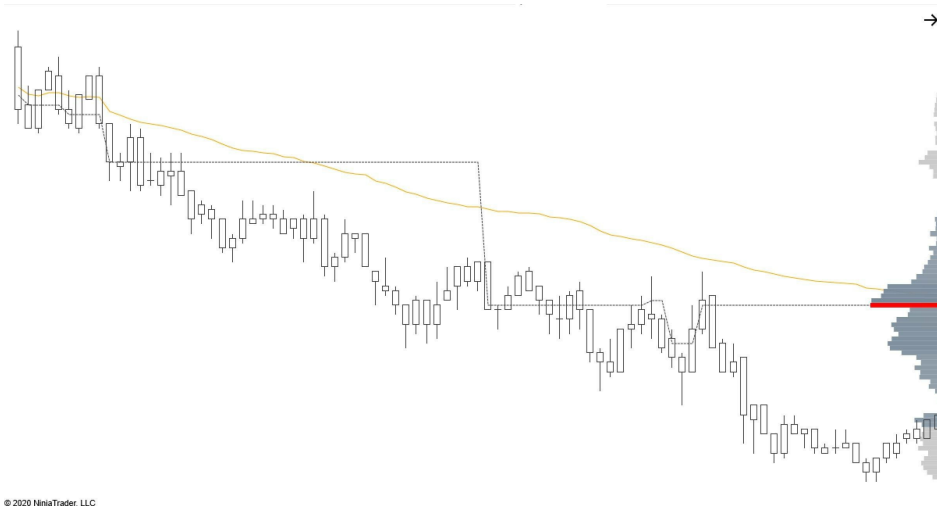
If in addition to this we have the possibility to evaluate the relationship between VPOC, VWAP and price, it will be one more footprint that will add strength to the analysis.

A relatively close VWAP and VPOC is a footprint that confirms the overall balance of the market. Possibly, price is in an extremely tight range and the only trading approach here would be to look for reversal at the extremes.



That is exactly what is happening on this day. The dark dotted line is the VPOC of the developing session and the orange dynamic line corresponds to the VWAP. Until a final imbalance to the downside occurs both levels remain relatively close and this generates constant fluctuations between them, causing a range day.

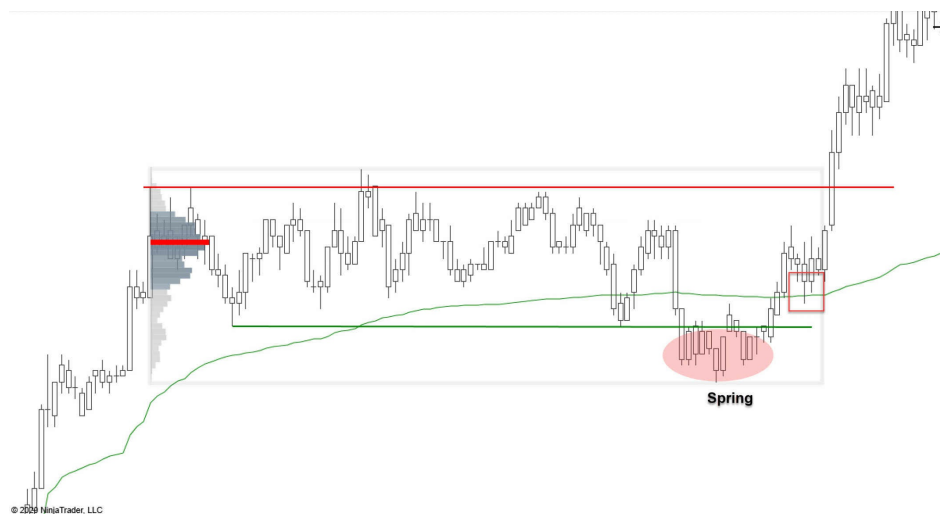
To determine an **imbalance** in favor of buyers we want to see the price above both the VWAP and the VPOC; while for control in favor of sellers we want to see the price below both levels.



In this chart we see an example of clear bearish control where at all

times of the session the price is below both the VWAP and the VPOC, acting as resistance to originate new bearish movements.

The temporality to use in these operative levels, as everything, will depend on the trader. For an intraday trader the most advisable will be to use the levels of the previous session and of the current session. Longer term traders may find it more useful to use such levels on a weekly basis (weekly VPOC and weekly VWAP). Particularly being a structure trader, I find it useful to take into account the weekly VWAP in conjunction with the VPOC of the structure, thus eliminating the seasonality. It is a matter of taste and this configuration should be adapted to the trading style of each trader.



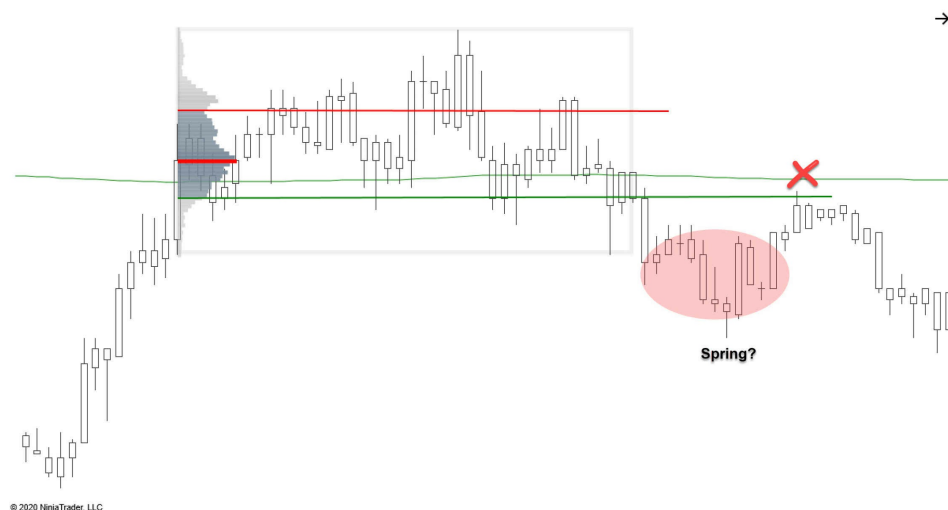
In this chart where we are analyzing a structure we could pull a profile of it and add the weekly VWAP (green dynamic line). It is a very good example to highlight the importance of the context above any other element.

If we see a potential Spring that manages to position itself above the VWAP we would be ready to look for a long entry. But if we look closely, the VPOC of the profile we would still be against the VWAP that

is marked. We are suggesting the possibility of trading in favor of the more operating levels the better, what do we do then? Context should prevail in cases like this. We know that VPOC is a very important balancing level, but we also know that the market will initiate an imbalance sooner or later; and that shakeout from below suggests to us the potential initiation of such an imbalance from the upside. In such a context we should assign greater relevance to the development of the structure above the VPOC.

We would have reasoned very differently if the price after the potential Spring had not even had the ability to move above the VWAP. That lack of strength would have suggested to us some bearish control and even an entry in the opposite direction could have been priced in seeing Spring's labeled move as a genuine bearish breakout event. Any action must be confirmed or rejected with subsequent price action.

That is just what we have in this example. In a potential Spring situation the price attempts to re-enter the range but is unable to position above the extreme of the structure, the profile VAL and the weekly VWAP. This is a sign of significant weakness that would lead us to treat such a pattern as distributive.



When working with different levels it is also important to keep in mind that we can use several timeframes in conjunction, for example, two VWAPs in weekly and monthly timeframes. This configuration is really interesting if we decide to analyze a longer term context.

5.7.3 ANALYZING TREND HEALTH

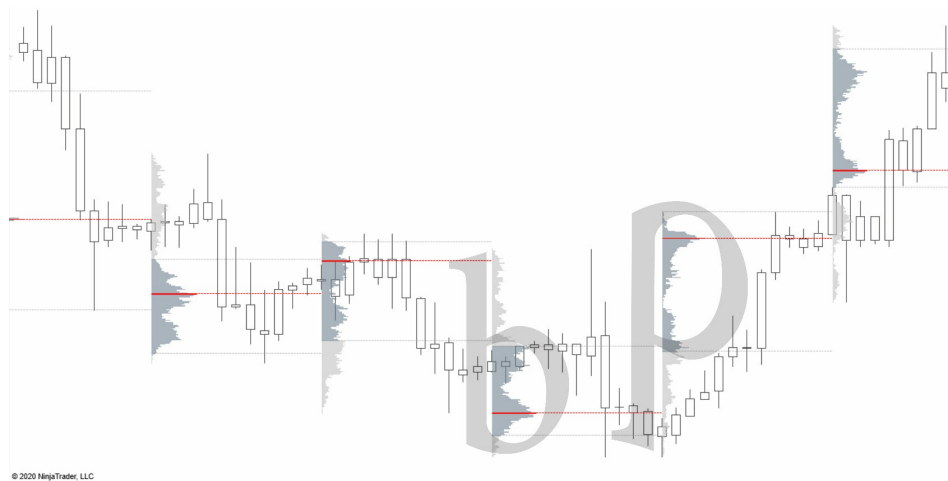
A very interesting footprint provided by the Volume Profile is the continuous analysis of the profile of the sessions. When we are observing a bullish movement, a symptom of the health of the movement would be to observe that the value areas (and therefore the VPOCs) of the sessions are generated higher and higher. This suggests that the value of the asset is being accepted in the new trading areas, it is reaching and therefore the trend is likely to continue. In this context of control by buyers, we should be looking for a pullback to trading zones predisposed to go long.

The same applies to downtrends. An unmistakable sign of its health would be to see that the value areas that leave the sessions are observed each time at lower levels, denoting acceptance of the price. In this scenario of bearish control, the most advisable thing to do is to identify areas of potential resistance to look for short positions.

Something that would alert us to the health of this movement would be to observe overlapping between several areas of value as well as to see that some of them are moving against the direction of the trend, losing the dynamics it had been carrying. We would already be observing a consolidation of the price and it could alert us of a change of character.

By Wyckoff methodology we could surely already identify a lateralization process and it would be interesting to start analyzing the traces of this structure in order to try to determine in which direction the next imbalance will occur.

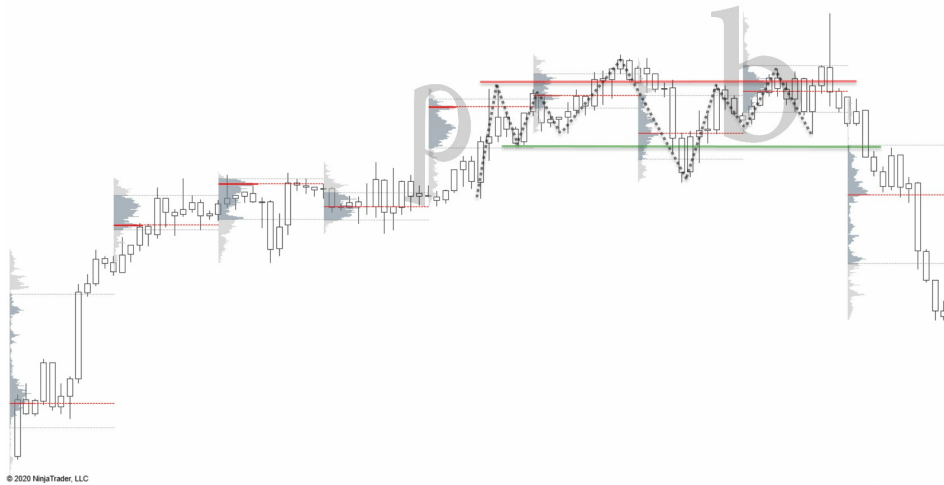
This potential change in the perception of value is very clearly visualized with the P and b patterns. If the market is in the middle of a bearish trend leaving the value zones lower and lower and then suddenly a bP pattern appears, this could be the end of the bearish movement or at least a temporary halt. The fact that the market has developed this P pattern after the b pattern suggests a change in the perception of value. At least temporarily it no longer wants to trade at lower prices and we could even be facing the beginning of an uptrend.



Rather than seeking to clearly observe the ideal pattern, what is interesting is the evolution of the value generation, i.e. the rotation of the Value Areas. In this example the participants have had enough capacity to generate value above the previous one and this alone should alert us to the health of the bearish movement and even put us on alert for a possible accumulation.

The same would be true in reverse. If we are looking for a bearish

market turn, a signal that would add strength to such analysis would be to observe a Pb pattern.



This example is a bit more complex and interesting to analyze. The first thing that is striking is that the pattern does not occur in tandem, but rather two sessions in between. On the other hand, we can see how profile b does not generate value below the Value Area of the profile

P. This is a real example; the market behaves erratically most of the time so looking for the appearance of perfect patterns can be a waste of time. It would seem more interesting to stay with the dynamics within them; what they implicitly suggest to us.

After the appearance of the P-type profile there is a change in character and the market begins a rotation process. The following two days the value areas are overlapped, showing the acceptance of these price levels by the vast majority of participants. Finally, the creation of the type b profile triggers the bearish imbalance. At that precise moment when the price is below the value area of profile b and after seeing all that has been analyzed previously, a short scenario could be proposed, seeking the effect of this potential distributive structure.

If we look closely at b, the imbalance does not occur at the beginning of the session as theory suggests, but rather in the last part of the session. The key is that this imbalance is rejected and re-enters the value area. In essence this is the implication behind the pattern: it makes no difference when the imbalance occurs as long as the price generates a rejection and re-enters the value area. For trading purposes what would be less interesting for us would be to observe the imbalance at the end of the session and whose close is established outside the value area.

Although the ideal in potential distributive schemes would be to observe first the rejection to continue rising plus the generation of value below the previous sessions, the appearance of this protocol in reverse (first the generation of value below and then the rejection to trade at higher prices) implicitly suggests the same reading of change in the perception of value.

In the end, all cumulative and distributive schemes implicitly carry this change in the perception of value; and to a greater or lesser extent these P and b turn patterns will always be visualized.

- ❖ A bearish turn with pattern Pb is nothing more than a distribution that will have a greater or lesser duration which has been confirmed with the generation of value in b and it is possible that lower prices will follow.
- ❖ A bullish turn with bP pattern tacitly is an accumulation whose change in the perception of value possibly gives rise to higher prices.

5.7.4 VPOC MIGRATION

The current level of the VPOC represents agreement by both parties on the value of the asset, but

What reading to do in the event of a VPOC migration?

This issue has brought many traders upside down as the reading it offers has two different points of view. On the one hand many advocate that this is an unequivocal sign of the health of the movement and therefore suggests continuity in the direction of the trend. Many others advocate the position that a turn in the market is possibly originating.

The only objective is that it represents a value zone where price has been accepted due to the high trading it has generated. The question would be to determine the meaning of this value migration, whether as a continuation or as a reversal.

Under the principle that every market action must be confirmed or rejected by its subsequent reaction, the key lies in evaluating the subsequent price action following the checkpoint migration. As a general rule, if we do not see a continuation in the direction of the preceding move without consuming too much time we should question the health of that move.

TRADING WITH VPOC MIGRATIONS

Since we cannot know in advance whether a VPOC migration will